Teaching for Impact

Designing Effective & Open Training

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1. Teaching for Impact: Designing Effective & Open Training: Training Manual & Workbook

1.1 Project team

1.1.1 Talarify

Anelda van der Walt

Talarify is a South African consulting and training organisation specialising in capacity and community development in Open Science and Digital and Computational Research.

1.1.2 Prelude

Sheena O'Connell

Prelude is a South African consulting and training organisation specialising in professional skills development in Python, Teamwork and Education. Prelude also builds edTech products.

1.2 About this Workbook

This workbook was developed as a reference for a course offered to grantees of the Failure Modes of Engineering (FeME) Seed Funding instrument: Data for Change Training Fund in 2026.

This workbook is both a **training manual** and a **practical working document** to support the design and improvement of training.

It is intended for practitioners designing and delivering training in complex, resource-constrained, and context-specific environments.

1.3 Course Structure

The course consist of the following elements:

Module 1: Workshop

- **Aim**: to introduce participants to the theory around evidence-based teaching and OER development.
- **Time estimate and themes**: (total 8 hours teaching time)
- Lesson 1: Evidence-Informed Teaching in Practice - 4 hours
- Lesson 2: Developing accessible OERs - 4 hours
- This module is delivered online and recorded for participants to reference later.
- The training alternates between teaching, Q\&A, and tailored activities to help participants relate what they've learned to their own training proposal activities.

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Module 2: Discussion Sessions

- **Aim**: To provide participants with opportunities to consider what they've learned in Module 1 in the context of their own plans for teaching and content development through peer-learning and discussion.
- **Time estimate and themes**: (total 4.5 hours discussion time)
- Discussion 1: Applying evidence-based teaching practices in my training - 1.5 hours
- Discussion 2: Developing my content as an accessible OER - 1.5 hours
- Discussion 3: Applying the skills I've learned to teach my own material - 1.5 hours

Module 3: One-on-One Mentoring

- **Aim**: To facilitate individual conversations where participants practice their teaching, get feedback on content, and discuss and brainstorm unique ideas, challenges, and contexts with a mentor.
- **Time estimate:** 1 hour per participant.

1.4 How to Use the Workbook

The contents of this workbook were designed to be used:

1. as self-study material (in this case, we strongly recommend further reading using the reference materials);
2. as a training manual and workbook complementing the interactive workshops run by facilitators with experience in evidence-based teaching and open science;
3. as a **practical working document**, where you actively design, test, and refine your own training materials.

> **Tip**
>
> Use the templates and activities throughout the workbook to develop your own materials as you progress.

1.5 Expected Learning Outcomes

After working through this book, learners will be able to:

- design training aligned with real-world outcomes and constraints
- analyse learners, context, and systems to inform design decisions
- create participatory learning experiences that support application
- integrate practice, feedback, and evaluation into training
- decide when to reuse, adapt, or create materials
- prepare and share training materials as reusable Open Educational Resources

1.6 Keeping the Information Current

The information in this document reflects and references, to the best of the authors' knowledge, the status quo, recommendations, and resources at publication.

Should you have any suggestions for updating or extending the content, please contact info@talarify.co.za. Contributors will be acknowledged appropriately in future releases. All versions will be hosted on Zenodo.

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1.7 Re-using, Remixing, Reproducing

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1.8 Citation

TODO

1.9 Funding

The training and development of this workbook and associated resources were funded through the Failure Modes of Engineering (FeME) project.

1.10 Acknowledgements

This workbook and training were inspired by the Instructor Training material developed by The Carpentries and the Creative Commons Certificate Course.

1.11 AI Usage

Generative AI tools were used extensively to support drafting, restructuring, and refining content throughout this workbook. The author defined the overall structure, pedagogical approach, and conceptual framing, and iteratively guided and edited all AI-generated content.

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2. Introduction

This workbook is designed to shift how you think about training—from delivering content to enabling change in real-world contexts.

It is a practical guide for designing training that leads to meaningful outcomes. It is intended for practitioners working in complex, resource-constrained, and context-specific environments where training must be relevant, usable, and impactful.

Training is often approached as a one-time event. This workbook takes a different view. It treats training as an **intentional intervention within a broader system**, shaped by learners, context, constraints, and opportunities.

It also recognises that training rarely ends with a single delivery. What learners do afterwards—and how materials are reused, adapted, and improved—shapes longer-term impact.

2.1 How this workbook is structured

This workbook progresses from:

- designing training for real-world impact
- to preparing materials for reuse and adaptation
- to sharing and improving them over time as Open Educational Resources

Each section builds on the previous one and can also be revisited independently.

2.2 How to use this workbook

You can use this workbook:

- independently, as a step-by-step guide
- alongside a training programme or workshop
- as a working document to design or improve your own training

Throughout the workbook, you will find:

- practical steps and decision points
- examples grounded in real contexts
- templates to apply directly to your own work

> **Tip**
>
> Engage actively with the workbook. Apply each section to your own context as you progress.

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2.3 Approach and principles

This workbook is built on a set of practical principles:

- **Training is contextual**
  Design must reflect real-world conditions, not ideal scenarios.
- **Training is participatory**
  Co-design improves relevance, trust, and usability.
- **Learning requires action**
  Practice, feedback, and iteration are essential.
- **Reuse benefits you and others**
  Designing for reuse improves efficiency and sustainability.
- **Improvement is ongoing**
  Training and materials evolve through use, feedback, and adaptation over time.
- **Impact extends beyond the room**
  Your training influences not only participants, but also the people and contexts they work in.

2.4 What you will be able to do

By working through this workbook, you will be able to:

- design training aligned with real-world outcomes
- understand learners and context
- create participatory learning experiences
- integrate practice, feedback, and evaluation
- reuse, adapt, and share materials effectively
- improve your training and materials over time

Info

The aim is not perfection, but progress—designing something that works, improving it through use, and building from there.

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3. Part 1 - Designing Training

3.1 PART 1: Designing Training for Real-World Impact

3.1.1 1. Getting Started: Training as Intervention and System

3.1.2 2. Understanding Your Learners, Contexts, and Knowledge Systems

3.1.3 3. How People Learn: Cognitive, Social, and Contextual Perspectives

3.1.4 4. Defining Impact, Learning Outcomes, and Alignment

3.1.5 5. Co-design and Constraints

3.1.6 6. Designing Learning Experiences for Participation and Co-Creation

3.1.7 7. Practice, Feedback, and Iteration in Community Contexts

3.1.8 8. Assessment and Evaluation for Learning and Impact

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3.2 1. Getting Started: Training as Intervention and System

🎯 Learning Outcomes	📍 Guiding Questions
• Describe training as an intervention within a broader system • Identify key elements of the system your training operates in (actors, resources, constraints, influences) • Articulate a simple theory of change linking activities to outcomes and impact • Analyse how context, power, and positionality influence training design and outcomes • Identify key assumptions and risks that may affect the success of your training	• What system does your training sit within, and who are the key actors? • What change are you trying to contribute to, and how will training support that? • What assumptions are you making about how change will happen? • What factors outside your control could affect success? • How do your role and positionality influence the design and delivery of this training?

3.2.1 Why this matters

Many training programmes are designed as events:

- a workshop
- a course
- a set of materials

But training does not exist in isolation.

Training is an intervention in a system. It interacts with institutions, communities, resources, and constraints. Its outcomes depend on much more than what happens during a session.

If you design training without understanding the system it sits in, you risk:

- overestimating impact
- ignoring constraints
- designing something that cannot work in practice

> **This section helps you move from...**
>
> *"What should I teach?"*
> to:
> *"What change is possible in this system—and how can training contribute?"*

3.2.2 Core concepts

> **Training as an intervention**
>
> Training is a deliberate action intended to contribute to change, but it only works in relation to the wider system around it.

Training may introduce new knowledge, build skills, create opportunities for practice, or support changes in behaviour. But training alone rarely creates impact. Impact depends on whether learners can apply what they learn, whether the environment supports change, and how well the training is aligned with real needs.

Systems thinking (practical)
A system includes people, resources, relationships, constraints, and external influences. Your training sits within a broader system of people, conditions, and constraints. It does not control it.

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Theory of Change
A Theory of Change is a simple explanation of how your training is expected to lead to outcomes and impact:

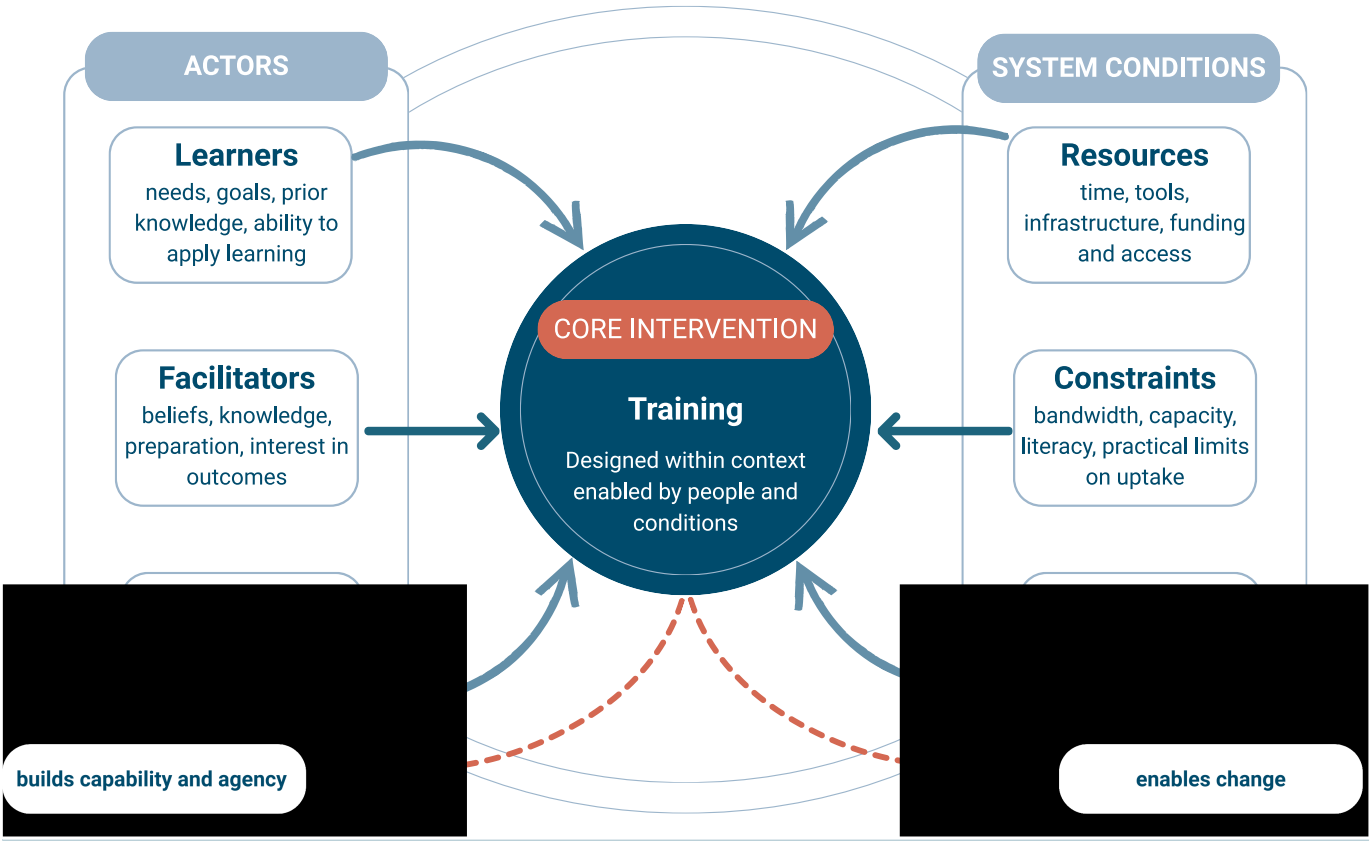
- Inputs → Activities → Outputs → Outcomes → Impact

It helps you make assumptions visible and identify where things may break down.

This model focuses on the *system-level logic* of change: how your training contributes to broader outcomes and impact.

In later sections, you will apply this same logic at the level of training design using a practical approach called backward design.

Positionality, roles, and power
Training is not neutral. Your role, background, authority, and assumptions shape what is prioritised, whose knowledge is valued, and how participants engage. You may be acting as a facilitator, co-learner, co-designer, or connector. Each role creates different possibilities.



3.2.3 Practical guidance

Step 1 — Map your system

Identify:

- who is involved
- what resources exist or are missing
- what constraints matter most
- what external factors influence success

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Step 2 — Define intended change

Be specific:

- What should participants do differently?
- What change do you hope the training contributes to?

Step 3 — Build a simple Theory of Change

Link:

- what you do
- what participants gain
- what changes afterwards

Keep it realistic.

Step 4 — Identify assumptions and risks

Ask:

- What must be true for this training to work?
- What is outside your control?
- What could prevent learners from applying what they learn?

Step 5 — Reflect on your role

Decide:

- Are you delivering, facilitating, or co-creating?
- How much influence will participants have?
- What power dynamics matter here?

Decision point

What level of change is realistic in your context?

- Individual skills only
- Changes in practice
- Institutional or community impact

Your answer shapes how you design the training.

3.2.4 Example

- **Context:** A team is designing climate data training for community organisers in a low-resource setting.
- **Decision:** Should they focus on advanced tools or practical low-tech approaches, and how much should participants shape the design?
- **Action:** They map the system and identify limited internet access, varied technical skills, and strong local knowledge. They decide to co-design parts of the training, use offline tools, and include locally relevant examples. They also shift their role from instructor to facilitator.
- **Outcome:** Participants are able to apply what they learn immediately, and the training can be adapted for future use in similar contexts.

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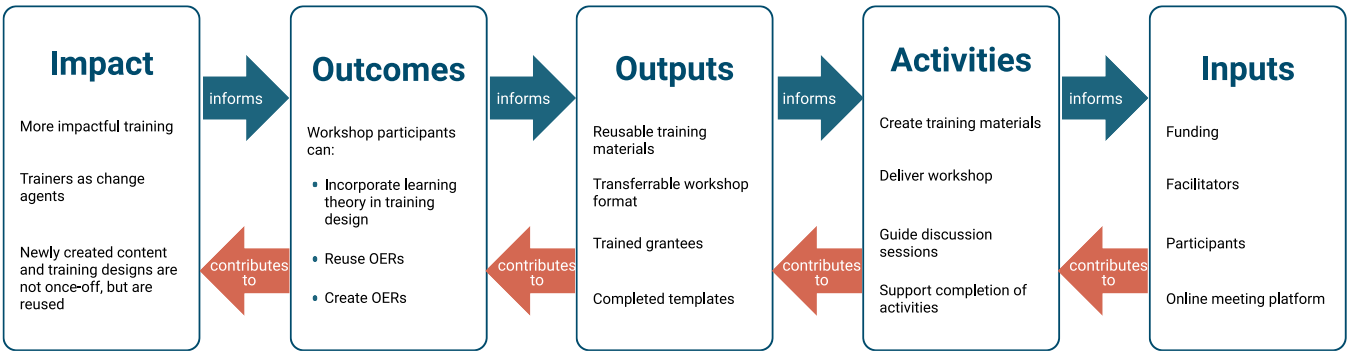

3.2.5 In practice

👉 Use Template 1: System Map

Include:

- **what to do:** Map actors, resources, constraints, relationships, and external influences
- **expected output:** A simple system map showing how your training fits into a broader context
- **approximate time:** 20–30 minutes

👉 Use Template 2: Theory of Change

Include:

- **what to do:** Link inputs, activities, outputs, outcomes, and impact
- **expected output:** A one-page Theory of Change for your training
- **approximate time:** 20–30 minutes

3.2.6 Key takeaways

Key takeaway

Training works within systems. Its success depends on context, not only content.

Key takeaway

A simple Theory of Change helps you design more realistically and identify key assumptions early.

3.2.7 Before you move on

You should now have:

- a basic system map of your training context
- a draft Theory of Change
- a clearer understanding of your role, assumptions, and key constraints

3.2.8 Further reading (optional)

- Meadows, D. (2008) — *Thinking in Systems: A Primer*
  → Supports: systems thinking and understanding training as part of a broader system

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→ Why it matters: provides practical tools for mapping actors, constraints, and system dynamics in training design
→ Source: https://donellameadows.org/archives/thinking-in-systems-a-primer/

- Weiss, C. (1995) — *Nothing as Practical as Good Theory: Exploring Theory-Based Evaluation for Comprehensive Community Initiatives*
  → Supports: Theory of Change linking activities to outcomes and impact
  → Why it matters: explains how making assumptions explicit improves programme design and evaluation
  → Source: https://www.jstor.org/stable/10.2307/1349159
- Freire, P. (1970) — *Pedagogy of the Oppressed*
  → Supports: positionality, power, and non-neutrality in training
  → Why it matters: highlights how power relations shape participation and knowledge in learning environments
  → Source: https://www.penguinrandomhouse.com/books/112874/pedagogy-of-the-oppressed-by-paulo-freire/
- Center for Theory of Change → Supports: background, benefits, origin and rationale of Theory of Change → Why it matters: mastering the theory of change can be very powerful for programme and project development → Source: https://www.theoryofchange.org/

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3.3 2. Understanding Your Learners, Contexts, and Knowledge Systems

🎯 Learning Outcomes	📍 Guiding Questions
• Analyse the characteristics, needs, and realities of your primary and secondary audiences • Identify contextual factors (access, time, language, culture, power) that influence participation and learning • Conduct a simple, participatory needs assessment to inform training design • Recognise and incorporate learners' prior knowledge, lived experience, and mental models • Identify potential barriers to participation and design strategies to reduce them	• Who are your learners, and what distinguishes primary from secondary audiences? • What constraints shape their participation? • What do learners already know? • What barriers might affect participation and success? • How can you incorporate lived experience and local knowledge?

3.3.1 Why this matters

In the previous section, you mapped your training as an intervention within a broader system.

This section focuses on one of the most important parts of that system: your learners.

Training is most effective when it aligns with how people actually live, work, and learn. Well-designed training does not start with content. It starts with understanding:

- who your learners are
- what shapes their participation
- what knowledge they already bring

When you design with learners in mind, your training becomes more relevant, easier to apply, and more likely to contribute to meaningful change.

This section helps you move from...

"What should I teach?"
to:
"Who are my learners, and what is possible in their context?"

3.3.2 Core concepts

Learners, context, and prior knowledge

Effective training design depends on understanding who your learners are, what conditions shape their participation, and what knowledge they already bring.

Primary and secondary audiences
Primary audiences are the direct participants in your training. Secondary audiences are the people they later support, train, or influence. If your participants act as intermediaries, your design must account for both immediate learning and downstream use.

Needs assessment
A needs assessment is a practical way to understand what learners need to do, what they already know, and what constraints shape that. It does not need to be complex. Short conversations, quick surveys, interviews, and co-mapping exercises are often enough.

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Context shapes what is possible in learning
Access, time, language, culture, social expectations, and power dynamics are not side issues. They are design conditions.

Lived experience and mental models
Learners bring prior knowledge, lived experience, habits, assumptions, and local expertise. New learning makes more sense when it connects to what already exists.

3.3.3 Practical guidance

Step 1 — Identify your audiences

Clarify:

- who your primary participants are
- who the secondary audiences are
- what responsibilities participants carry beyond the training

Step 2 — Understand learner realities

Gather information about:

- access to devices, internet, tools, and space
- time constraints and competing priorities
- language and communication preferences
- social and cultural context
- what success would look like from the learners' perspective

Step 3 — Conduct a simple needs assessment

Use:

- short conversations
- informal interviews
- quick surveys
- co-mapping exercises

Focus on:

- what learners already know
- what they need to do
- what challenges they face
- what strengths they already have

Step 4 — Identify barriers and opportunities

Ask:

- What might prevent participation or success?
- What can you build on?
- What existing knowledge or practice can become part of the training?

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Step 5 — Design with learners, not only for them

Where possible:

- involve learners in shaping content
- use examples from their context
- validate different forms of knowledge
- avoid assuming that expert knowledge is the only knowledge that matters

Decision point

How much do you need to adapt your training to fit your learners' realities?

- Minimal adaptation
- Moderate adaptation
- Significant redesign

Your answer will shape your content, format, pace, and delivery choices.

3.3.4 Example

- **Context:** A team is designing training for early-career researchers who will later train community members on climate data tools.
- **Decision:** How should the training account for both researchers and community members?
- **Action:** They conduct a simple needs assessment and learn that researchers are comfortable with technical tools, while community members have limited access and prefer practical examples. They design training that prepares researchers to adapt content, includes low-tech alternatives, and emphasises facilitation skills.
- **Outcome:** The training becomes more realistic, more transferable, and more useful for both the immediate and secondary audiences.

3.3.5 In practice

👉 Use Template 3: Learner Reality Mapping

Include:

- **what to do:** Map your audiences, their realities, constraints, strengths, and needs
- **expected output:** A structured summary of learner needs and contextual design considerations
- **approximate time:** 20–30 minutes

3.3.6 Key takeaways

🔥 **Key takeaway**

Good training begins with learner realities, not assumptions.

🔥 **Key takeaway**

Prior knowledge and lived experience are assets. Design should connect to them, not work against them.

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3.3.7 Before you move on

You should now have:

- a clearer picture of your primary and secondary audiences
- a basic needs assessment
- a list of learner realities, barriers, strengths, and design implications

3.3.8 Further reading (optional)

- Knowles, M., Holton, E., & Swanson, R. (2015) — *The Adult Learner*
  → Supports: understanding learner needs, prior knowledge, and self-directed learning
  → Why it matters: provides foundational principles for designing training aligned with adult learners' realities
  → Source: https://www.routledge.com/The-Adult-Learner-The-Definitive-Classic-in-Adult-Education-and-Human/Knowles-Holton-Swanson/p/book/9781138209398
- Kolb, D. (1984) — *Experiential Learning: Experience as the Source of Learning and Development*
  → Supports: connecting learning to lived experience and prior knowledge
  → Why it matters: explains how experience shapes learning and informs participatory training design
  → Source: https://www.pearson.com/en-us/subject-catalog/p/experiential-learning-experience-as-the-source-of-learning-and-development/P200000003495/9780132952613
- Merriam, S., & Bierema, L. (2014) — *Adult Learning: Linking Theory and Practice*
  → Supports: needs assessment and contextual influences on learning
  → Why it matters: integrates theory and practice for designing context-sensitive and learner-centred training
  → Source: https://www.wiley.com/en-us/Adult+Learning%3A+Linking+Theory+and+Practice-p-9781118130575

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3.4 3. How People Learn: Cognitive, Social, and Contextual Perspectives

🎯 Learning Outcomes	📍 Guiding Questions
• Explain key principles of how people learn (cognitive load, retrieval, spacing, social learning) • Identify factors that support or hinder effective learning • Apply strategies to reduce cognitive overload and improve retention • Design learning experiences that include retrieval, practice, and reinforcement • Incorporate social and contextual dimensions of learning into training design	• How do people move information from short-term to long-term memory? • Where might cognitive overload occur in your training? • How will learners actively retrieve and apply knowledge? • How can learning be reinforced over time? • How will you incorporate social learning and collaboration?

3.4.1 Why this matters

In the previous section, you analysed your learners, their contexts, and the knowledge they bring. This section focuses on how learning actually happens so that your design decisions support it.

Training is not effective simply because content is delivered. Content supports learning, but learning depends on what learners do with it.

Learning is more likely to happen when participants:

- engage actively
- connect new ideas to what they already know
- retrieve and apply knowledge
- revisit key ideas over time
- learn with and from others

If you design without understanding how learning works, you risk overwhelming learners, creating passive experiences, and limiting retention and real-world application.

> **This section helps you move from...**
>
> *"How do I present content?"*
> to:
> *"How do people actually learn—and how do I design for that?"*

3.4.2 Core concepts

> **How learning strengthens**
>
> Learning becomes more durable when cognitive load is managed, learners actively retrieve and apply knowledge, key ideas are revisited over time, and interaction supports sense-making.

Cognitive load
Working memory is limited. When too much new information appears at once, learners are more likely to struggle to process it. Clear instructions, smaller chunks, and simpler materials reduce unnecessary load.

Retrieval and application
Learning becomes more robust when people recall information, explain it, use it, and make decisions with it. Re-reading alone is rarely enough.

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Spacing and reinforcement
Single exposure is usually insufficient. Revisiting key ideas across sessions and in new contexts improves retention.

Social learning
Discussion, collaboration, observation, and shared problem-solving help learners test understanding and refine ideas.

3.4.3 Practical guidance

Step 1 — Reduce unnecessary cognitive load

- break content into smaller segments
- keep instructions clear and simple
- avoid introducing multiple new ideas at once
- remove avoidable complexity

Step 2 — Design for active learning

For each part of your training, ask:

- What are learners doing?
- What will they produce, decide, explain, or apply?

Include:

- short exercises
- problem-solving tasks
- explanation and reflection activities

Step 3 — Build in retrieval

Design opportunities where learners:

- recall key ideas
- answer questions without prompts
- apply concepts in realistic tasks

Step 4 — Plan for reinforcement

- revisit key ideas across sessions
- use follow-up activities
- encourage continued practice beyond the session

Step 5 — Use social learning intentionally

- include discussion prompts
- design small-group or peer activities
- create opportunities for learners to compare approaches and explain their thinking

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Decision point

How much active learning and retrieval will you build into your training?

- Mostly content delivery with limited activity
- A balanced mix of explanation and practice
- Practice-focused, highly interactive design

Your choice will influence retention, confidence, and transfer into practice.

3.4.4 Example

- **Context:** A team is developing training on local climate risk assessment.
- **Decision:** Should they cover more content, or reduce content to create time for retrieval and practice?
- **Action:** They cut non-essential slides, break the material into smaller chunks, add short recall questions, revisit key ideas later in the session, and use pair discussion to test understanding.
- **Outcome:** Participants tend to remember more, ask better questions, and are better able to apply the method afterwards.

3.4.5 In practice

👉 Use Template 4: Learner Experience Audit

Include:

- **what to do:** Analyse one part of your training to identify cognitive overload, passive learning, and missed opportunities for retrieval, practice, and interaction
- **expected output:** A short diagnostic identifying key learning risks and 2–3 concrete improvements to strengthen learning
- **approximate time:** 15–20 minutes

👉 Revisit Template 3: Learner Reality Mapping

Include:

- **what to do:** Use your learner realities to identify where overload, confusion, or disengagement may occur
- **expected output:** Notes on how learner constraints affect learning design decisions
- **approximate time:** 10–15 minutes

3.4.6 Key takeaways

🔥 **Key takeaway**

Learning is strengthened by retrieval, practice, reinforcement, and interaction, not only by explanation.

🔥 **Key takeaway**

Reducing unnecessary complexity creates more room for learners to think, apply, and remember.

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3.4.7 Before you move on

You should now have:

- identified where overload may occur in your design
- selected ways to build retrieval and reinforcement into learning
- a clearer plan for how learners will actively engage with content

3.4.8 Further reading (optional)

- Sweller, J. (1988) — *Cognitive Load During Problem Solving: Effects on Learning*
  → Supports: cognitive load and managing working memory in training design
  → Why it matters: explains why reducing overload improves learning effectiveness
  → Source: https://doi.org/10.1207/s15516709cog1202_4
- Brown, P., Roediger, H., & McDaniel, M. (2014) — *Make It Stick: The Science of Successful Learning*
  → Supports: retrieval, spacing, and reinforcement strategies
  → Why it matters: synthesises evidence-based techniques for improving retention and transfer
  → Source: https://www.hup.harvard.edu/books/9780674729018
- Bandura, A. (1977) — *Social Learning Theory*
  → Supports: social learning through observation and interaction
  → Why it matters: explains how collaboration and modelling support learning processes
  → Source: https://psycnet.apa.org/record/1977-25733-000

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3.5 4. Defining Impact, Learning Outcomes, and Alignment

🎯 Learning Outcomes	📍 Guiding Questions
• Define clear, observable, and actionable learning outcomes	• What should learners be able to do as a result of this training?
• Apply backward design to link learning outcomes, evidence, and activities	• What will learners need to produce, do, or decide to show they have achieved the intended learning outcomes?
• Design activities that enable learners to practise and demonstrate learning outcomes	• Do your planned activities directly support your intended learning outcomes?
• Develop supporting content that directly enables successful participation in activities	• What content do learners need to successfully participate in the activities?
• Diagnose and correct misalignment between learning outcomes, activities, and assessment	• Where might misalignment exist between learning outcomes, activities, assessment, and content?

3.5.1 Why this matters

In the previous section, you explored how people learn and what supports effective learning.

This section builds on that by structuring your training design so that learning can lead to clear, observable outcomes.

Training impact is often not achieved not because of poor delivery, but because:

- learning outcomes are unclear
- activities do not support what learners need to do
- success is not defined meaningfully
- content does not support the expected learning

As a result, participants may engage with training but struggle to apply what they learned in practice.

> **This section helps you move from...**
>
> *"What should I include?"*
> to:
> *"What should learners be able to do—and how will I support and recognise that?"*

3.5.2 Core concepts

> **Learning outcomes and alignment**
>
> Effective training starts with clear learning outcomes, defines evidence of success, and then designs activities and assessment that support those outcomes.

Outputs, outcomes, and impact (from Theory of Change)
Outputs are immediate products of training, such as worksheets, discussions, or completed tasks. Outcomes are what learners do differently after the training in their real-world context. Impact is the longer-term change that results.

In this section, we focus specifically on **learning outcomes**—what participants should be able to do by the end of the training—which contribute to these broader outcomes.

Backward design
Building on the theory of change, backward design applies the same logic at the level of training design.

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Instead of starting with content, start with what learners should be able to do, then define how you will recognise that, and then design activities to support it.

Backward design follows a simple sequence:

- Learning outcomes → Evidence → Activities

Evidence helps you judge whether learners have achieved the intended learning outcomes. This can be captured in different ways, including assessment, where learners demonstrate what they can do. You will develop assessment in more detail in later sections.

Content is developed after activities have been designed, to support those activities.

It keeps design focused on what matters.

Constructive alignment
Alignment means your learning outcomes, activities, and assessment support each other. Learners should practise what they are expected to do, and assessment should reflect that same capability.

How the design elements fit together

Effective training design involves four key elements. This section focuses on how these elements fit together at a design level. You will develop each of these in more detail in later sections.

- **Learning outcomes** — what learners should be able to do by the end
- **Assessment (evidence)** — how you will recognise that learning has occurred
- **Activities** — how learners will practise and apply skills
- **Content** — the information and resources that support learning

These elements are connected, but they do not have equal roles. Activities are the bridge between learning outcomes and assessment: they are where learning is practised and made visible.

Design order (how you plan)

When designing training, follow this sequence:

1. **Define learning outcomes**
2. **Decide what evidence will demonstrate success**
3. **Design activities that enable learners to achieve those outcomes**
4. **Develop content that supports those activities**

This helps ensure that everything in your training serves a clear purpose.

Delivery order (what learners experience)

During the training, learners typically experience:

- content (input)
- activities (practice)
- assessment (demonstration)

This means the design process works in the opposite direction to the learning experience.

Key principle

Content should not drive training design.

Content supports activities, and activities enable learners to work toward the intended learning outcomes.

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3.5.3 Practical guidance

The steps below help you structure your training design. Later sections will support you in developing activities, content, and assessment in more detail.

Step 1 — Define intended learning outcomes

Strong learning outcomes are:

- observable
- action-oriented
- relevant to real-world use

For example:

- “Interpret climate data to identify local risks”
- “Use a tool to analyse and communicate findings”

You can use the Bloom’s Taxonomy — Quick Guide to help choose action verbs and vary the level of complexity (e.g. identify vs analyse vs create), but outcomes do not need to progress sequentially through the taxonomy—focus on what learners need to do in practice.

Step 2 — Define evidence of success

Ask:

- What would demonstrate that learners have achieved the intended learning outcomes?
- What would participants produce, do, or decide?

Keep this manageable. Two or three clear indicators are enough.

Step 3 — Design aligned activities

For each learning outcome, design activities where learners can:

- practise the skill
- apply knowledge
- receive feedback

Step 4 — Develop supporting content

Once activities are defined, identify and create only the content needed to support those activities.

This may include:

- explanations
- examples
- demonstrations
- resources

Avoid creating content that is not directly used in an activity.

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Step 5 — Check and refine alignment

Review and revise your design (this will be iterative across later sections):

- What is taught but not practised?
- What is practised but not assessed?
- Which activities do not contribute to learning outcomes?

Step 6 — Ensure contextual relevance

Make sure learning outcomes reflect:

- what learners actually need to do
- the conditions they work within
- the decisions they are likely to face

Decision point

How specific and practice-oriented will your learning outcomes be?

- Broad and general
- Moderately specific
- Clear, observable, and linked to practice

Your choice will affect the quality of your activities, assessment, and evaluation.

3.5.4 Example

- **Context:** A team wants to train participants on climate data interpretation.
- **Decision:** Should the main learning outcome be “understand climate data” or something more specific?
- **Action:** They rewrite the outcome as “interpret a local dataset to identify two practical risks and explain them to a non-specialist audience.” They then define evidence, design a matching activity, and remove unrelated content.
- **Outcome:** The training becomes clearer, more practical, and easier to assess.

3.5.5 In practice

👉 Use Template 5: Learning Outcomes

Include:

- **what to do:** Draft or revise outcomes so they are observable and action-oriented
- **expected output:** A short set of clear learning outcomes
- **approximate time:** 15–20 minutes
- **optional:** Use the Bloom’s Taxonomy — Quick Guide if you need help refining action verbs.

👉 Use Template 6: Alignment Table

This is an initial alignment check to structure your design. You will revisit and improve this as you develop activities and assessment in later sections.

Include:

- **what to do:** Create a first draft linking each learning outcome to evidence, activities, and assessment. You will refine this in later sections.
- **expected output:** A first draft of the alignment table
- **approximate time:** 20–30 minutes

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3.5.6 Key takeaways

Key takeaway

Learning outcomes should describe what learners will be able to do, not only what content will be covered.

Key takeaway

Alignment is practical: if an activity or assessment does not support a learning outcome, it probably does not belong.

3.5.7 Before you move on

You should now have:

- a clearer set of learning outcomes
- initial indicators or evidence of success
- a stronger link between outcomes, activities, and assessment

3.5.8 Further reading (optional)

- Biggs, J., & Tang, C. (2011) — *Teaching for Quality Learning at University*
  → Supports: constructive alignment between outcomes, activities, and assessment
  → Why it matters: provides a widely used framework for aligning teaching design decisions
  → Source: https://www.mheducation.com/highered/product/teaching-quality-learning-university-biggs/M9780335242757.html
- Wiggins, G., & McTighe, J. (2005) — *Understanding by Design*
  → Supports: backward design (outcomes → evidence → activities)
  → Why it matters: offers a practical model for outcome-driven training design
  → Source: https://www.ascd.org/books/understanding-by-design-expanded-2nd-edition-9781416600350
- Kirkpatrick, D., & Kirkpatrick, J. (2006) — *Evaluating Training Programs: The Four Levels*
  → Supports: distinguishing outputs, outcomes, and impact
  → Why it matters: provides a widely used model for evaluating training effectiveness
  → Source: https://www.berrett-koehler.com/product/evaluating-training-programs/

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3.6 5. Co-design and Constraints

🎯 Learning Outcomes	📍 Guiding Questions
• Use co-design principles to make more relevant and context-aware design decisions • Identify key constraints that shape what is possible in your training • Make informed trade-offs based on context, priorities, and available resources • Create opportunities for meaningful participant contribution in your design • Avoid extractive or tokenistic participation practices	• Who is currently shaping your training, and who should be involved? • What constraints most strongly affect your design decisions? • What trade-offs are you making, and why? • How can participants meaningfully influence the training design? • How will you make sure participants have real influence, not just provide input?

3.6.1 Why this matters

You have already analysed your learners, their context, and what they bring to the training.

But understanding alone is often not enough.

The key question now is:

What do you do differently *because* of what you know?

Many training designs still tend to default to:

- ideal conditions
- fixed content
- one-way delivery

Even when the designer knows these don't match reality.

This is where co-design and constraints become especially important.

> **This section helps you move from...**
>
> *"I understand my learners and context"*
> to:
> *"I design differently because of that understanding"*

This lesson is not about re-analysing your learners or context.

Instead, it introduces **co-design and constraints** as a lens to apply when making design decisions.

You will use this lens to:

- adjust your learning outcomes if needed
- shape the kinds of activities you design
- decide how learners participate and contribute
- make realistic choices about format, time, and resources

In other words:

You are not starting from scratch — you are working with what you already know.

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3.6.2 Core concepts

Co-design under constraints

Time, access, infrastructure, language, institutional rules, social expectations, and power relations shape what is possible. These are design conditions, not just limitations.

Co-design
Participants can help make decisions about the training itself — such as what to focus on, how it is structured, and what activities are included.

Co-design vs co-creation

- **Co-design** = participants help decide **what the training will be and how it will work**
- **Co-creation** = participants contribute to **creating or shaping materials, ideas, or outputs**

In short:

- Co-design = **decision-making about the training**
- Co-creation = **making and shaping things together**

Examples:

- Co-design: participants help choose topics, define outcomes, or suggest how an activity should work
- Co-creation: participants generate examples, contribute content, or build outputs during activities

Both can happen at different stages, but they are different in purpose:

- Co-design shapes the training
- Co-creation shapes what is created within or for the training

Constraints
Time, access, infrastructure, language, institutional rules, social expectations, and power relations shape what is possible.

Trade-offs and ethical participation
Design decisions usually involve trade-offs between depth and breadth, participation and time, complexity and accessibility. Ethical co-design means participants have real influence and benefit from the process.

Not all co-creation is equal
Co-creation can vary in how open contributions are and who makes decisions.

- In some cases, participants have real influence over decisions
- In others, they contribute ideas but final decisions are made elsewhere

Be clear about what kind of participation you are offering, and avoid presenting limited input as shared decision-making.

3.6.3 Practical guidance

Step 1 — Identify who is shaping the training

Clarify:

- who is currently making key decisions
- who should be involved but is not yet included
- what decisions participants could meaningfully influence

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Step 2 — Identify key constraints

Map the constraints that matter most:

- time and scheduling
- technology and infrastructure
- materials and funding
- language and communication
- institutional or social dynamics
- anything else that is relevant to your specific context

Step 3 — Design with participants

Use manageable approaches such as:

- short pre-session consultations
- example gathering
- feedback on draft activities
- in-session adaptation based on participant input
- use participant input to surface constraints you may not have identified

Step 4 — Make trade-offs explicit

Ask:

- What matters most for the outcomes?
- What is essential?
- What can be simplified, deferred, or removed?

Step 5 — Ensure participation is meaningful

Check that participants:

- have real influence over something that matters
- are not only providing unpaid input with no visible effect
- see how their knowledge is used in the design

Decision point

How much influence will participants have in shaping your training?

- Minimal input
- Moderate input
- Strong co-design

Your choice will influence relevance, trust, and long-term usefulness.

3.6.4 Example

- **Context:** A team is designing training for local organisations with very different levels of digital access.
- **Decision:** Should they finalise the training before speaking to participants, or involve them first?
- **Action:** They run short pre-session conversations, ask participants what examples are most relevant, and identify where offline options are needed. They cut some planned content to make room for locally relevant material.

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- **Outcome:** The training becomes more realistic, more trusted, and easier for participants to use.

3.6.5 In practice

👉 Use Template 7: Co-Design Plan

Include:

- **what to do:** Identify who to involve, what decisions they can shape, and how input will be used
- **expected output:** A practical co-design plan
- **approximate time:** 20–30 minutes

👉 Revisit Template 1: System Map

Include:

- **what to do:** Add the key constraints and power dynamics affecting your training
- **expected output:** An updated system map with clearer design implications
- **approximate time:** 10–15 minutes

3.6.6 Key takeaways

> 🔥 **Key takeaway**
>
> Constraints are not design failures. They are part of the design brief.

> 🔥 **Key takeaway**
>
> Co-design improves relevance, trust, and usability when participants have real influence.

3.6.7 Before you move on

You should now have:

- identified who should shape your training
- mapped the key constraints affecting design choices
- made initial decisions about participation, trade-offs, and ethical engagement

3.6.8 Further reading (optional)

- Sanders, E., & Stappers, P. (2008) — *Co-creation and the New Landscapes of Design*
  → Supports: co-design principles and participant involvement
  → Why it matters: explains how involving stakeholders improves relevance and outcomes
  → Source: https://doi.org/10.1080/15710880701875068
- IDEO.org (2015) — *The Field Guide to Human-Centered Design*
  → Supports: designing with participants under real-world constraints
  → Why it matters: provides practical methods for participatory design in constrained contexts
  → Source: https://www.designkit.org/resources/1
- Chambers, R. (1994) — *Participatory Rural Appraisal (PRA): Challenges, Potentials and Paradigm*
  → Supports: ethical participation and avoiding extractive practices
  → Why it matters: offers Global South-informed approaches to inclusive and participatory design
  → Source: https://doi.org/10.1016/0305-750X(94)90030-2

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3.7 6. Designing Learning Experiences for Participation and Co-Creation

🎯 Learning Outcomes	📍 Guiding Questions
• Design activities where learners actively work toward the intended outcomes • Structure activities with clear instructions, materials, timing, and outputs • Design for inclusive participation so all learners are actively involved • Enable co-creation by drawing on learners' knowledge, experience, and context • Support learners through challenging parts of an activity using appropriate scaffolding • Adapt activities to fit your specific context (time, resources, group)	• What will learners actually do in this activity? • What do I need to prepare or provide so they can do it? • How will I make sure everyone participates, not just a few people? • How can learners contribute their own knowledge, experience, or ideas (co-creation)? • Where might learners get stuck, and how will I support them? • Will this work in my context (time, resources, group)?

3.7.1 Why this matters

In the previous section, you considered how co-design and constraints shape your design decisions.

Now, you apply that thinking to designing learning activities.

This section focuses on turning your outcomes into effective learning experiences where participants actively engage, contribute, and apply what they learn.

Activities are the bridge between your learning outcomes and what learners actually do in practice. They are where learning is practised and made visible.

Training often breaks down at the activity level:

- activities are unclear
- participation is limited
- tasks do not support outcomes
- activities do not fit the context

Effective training is strongly influenced by what learners do, not only by what you present.

> **This section helps you move from...**
>
> *"What content should I include?"*
> to:
> *"What should learners do to achieve the outcomes — in this context?"*

3.7.2 Core concepts

> **Structured, participatory activity design**
>
> Strong learning activities are clearly structured, actively engage learners, and provide enough support at the right time to move from guided participation to more independent practice.

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Activities as part of alignment
Activities are where learners work toward the intended outcomes and generate evidence of learning. They should be closely aligned with what learners need to do and what success looks like.

Co-creation in activities
In this section, co-creation refers specifically to how learners contribute within activities. Co-creation can also happen during design and material development, but here the focus is on how it supports learning during the training itself.

Activity structure
A well-structured activity includes:

- objective
- instructions
- materials
- time
- output
- feedback

Clarity reduces confusion and makes participation easier.

Active learning
Active learning means learners are discussing, solving, analysing, making, or applying something meaningful.

Scaffolding, sequencing, and psychological safety
Scaffolding provides support early and reduces it over time. Sequencing places simpler or more guided tasks before more complex ones. Psychological safety matters because learners participate more fully when they know they can contribute, try, and revise without embarrassment or unnecessary risk.

Learning is more likely to be effective when activities:

- focus attention on key ideas (managing cognitive load)
- require learners to recall, use, or apply knowledge (retrieval)
- connect to realistic tasks and contexts

3.7.3 Practical guidance

Step 1 — Design activities from outcomes

For each outcome, define:

- what learners will do
- what they will produce or decide
- how this supports the outcome

Step 2 — Structure each activity clearly

Specify:

- objective
- step-by-step instructions
- materials
- time
- output
- feedback

Write it so that another facilitator could run the activity.

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Step 3 — Design for participation

Design participation intentionally:

- use pair or small-group work
- allow think time before discussion
- offer multiple ways to contribute
- make expectations visible

Step 4 — Apply scaffolding and sequencing

Support learners by:

- starting with guided examples
- increasing complexity gradually
- reducing support over time
- sequencing activities so each one prepares for the next

Step 5 — Adapt activities using co-design and constraints

Ensure activities:

- work with the available tools and time
- use locally relevant examples
- leave room for participant knowledge and examples
- support inclusive participation

Decision point

How structured and participatory will your learning activities be?

- Loosely structured and facilitator-led
- Moderately structured with some participation
- Highly structured, participant-driven, and interactive

Your choice will influence clarity, engagement, and learning quality.

3.7.4 Example

- **Context:** A team is designing a session to help researchers analyse and interpret climate data.
- **Decision:** Should they use a lecture-based approach or a structured hands-on activity?
- **Action:** They design an activity with a clear objective, step-by-step instructions, a real local dataset, small-group work, a defined output, and time for feedback. They begin with a guided example and then move to more independent work.
- **Outcome:** Participants actively engage, apply the method, and are more able to use it independently after the session.

3.7.5 In practice

👉 Use Template 8: Learning Activity Design

Include:

- **what to do:** Design or refine one activity with clear structure (objective, instructions, materials, time, output, feedback)
- **expected output:** A well-structured activity aligned with your learning outcomes
- **approximate time:** 20–30 minutes

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👉 Revisit Template 9: Practice & Feedback Plan

Include:

- **what to do:** Identify where learners will practise, receive feedback, and improve within your activity
- **expected output:** A clearer link between your activity and learning processes
- **approximate time:** 10–15 minutes

Suggested process

Step 1 — Start from outcomes
Select one learning outcome.

Step 2 — Define the activity
Decide what learners will do and produce.

Step 3 — Structure clearly
Add instructions, materials, time, and output.

Step 4 — Add support
Include scaffolding and participation strategies.

Step 5 — Check alignment
Ensure the activity supports the intended outcome.

3.7.6 Key takeaways

🔥 **Key takeaway**

Activities are where learning is practised and made visible. They should align closely with your intended outcomes.

🔥 **Key takeaway**

Clear structure, scaffolding, and psychological safety make it easier for learners to engage and improve.

3.7.7 Before you move on

You have now designed what learners will do in your training.

In the next lesson, you will focus on how learners practise, receive feedback, and improve through iteration.

You should now have:

- at least one clearly structured learning activity
- a plan for participation and support
- stronger alignment between your activities, outcomes, and context

3.7.8 Further reading (optional)

- Prince, M. (2004) — *Does Active Learning Work? A Review of the Research*
  → Supports: active learning and participation in training design
  → Why it matters: provides evidence that active engagement improves learning outcomes
  → Source: https://doi.org/10.1002/j.2168-9830.2004.tb00809.x

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- Vygotsky, L. (1978) — *Mind in Society: The Development of Higher Psychological Processes*
  → Supports: scaffolding and social dimensions of learning
  → Why it matters: explains how guided support enables learners to progress toward independence
  → Source: https://www.hup.harvard.edu/books/9780674576292
- Ambrose, S., Bridges, M., DiPietro, M., Lovett, M., & Norman, M. (2010) — *How Learning Works: Seven Research-Based Principles for Smart Teaching*
  → Supports: structuring activities and supporting effective participation
  → Why it matters: translates learning science into practical guidance for designing effective activities
  → Source: https://www.wiley.com/en-us/How+Learning+Works%3A+Seven+Research-Based+Principles+for+Smart+Teaching-p-9780470484104

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3.8 7. Practice, Feedback, and Iteration in Community Contexts

🎯 Learning Outcomes	📍 Guiding Questions
• Design meaningful, outcome-aligned opportunities for practice • Plan feedback processes that include appropriate sources, timing, and format • Design feedback so that it is specific, timely, and actionable • Build iteration and revision cycles into learning activities • Facilitate peer learning and co-learning in inclusive community contexts	• Where will learners practise the key knowledge or skills? • How will practice connect directly to the intended outcomes? • What feedback will learners receive, from whom, and at what points? • How will feedback help learners improve while revision is still possible? • Where are revision and iteration built into the learning process? • How will participants learn with and from each other in this context?

3.8.1 Why this matters

In the previous section, you designed learning activities that support participation and engagement.

This section focuses on how those activities lead to meaningful learning through practice, feedback, and iteration.

Learning does not happen through exposure alone. Participants need:

- opportunities to practise
- feedback to improve
- time to revise and strengthen their work

Without this, even well-designed activities remain theoretical and difficult to apply.

> **This section helps you move from...**
>
> *"Did I deliver the activity?"*
> to:
> *"Did learners practise, receive feedback, and improve within the activity?"*

3.8.2 Core concepts

> **Practice, feedback, and iteration**
>
> Learning becomes more robust when learners apply skills in realistic tasks, receive useful feedback, and improve through repeated cycles of attempt, response, and revision.

Practice
Practice should align with outcomes, reflect real use, and include enough support to help learners make progress.

Feedback
Useful feedback is specific, timely, and actionable. It helps learners better understand what is working and what to improve next.

Iteration and productive challenge
Improvement typically happens through cycles of attempt, feedback, and revision. Learners do not need to get everything right on the first try. Productive challenge is part of learning when it is supported and followed by feedback.

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3.8.3 Practical guidance

Step 1 — Design meaningful practice

Design activities so learners:

- apply key skills
- work on realistic tasks
- connect practice directly to outcomes

Use a mix of:

- guided practice
- independent practice
- collaborative practice

Step 2 — Design feedback as part of the activity

Plan feedback as part of the activity:

- include checkpoints
- define what good performance looks like
- allow time for peer and facilitator feedback

Step 3 — Make feedback useful

Useful feedback typically:

- focus on specific aspects of performance
- include next steps
- arrive while revision is still possible

Step 4 — Build in iteration and versioning

Create opportunities for learners to revise and improve their work.

Simple approaches include:

- first attempt → feedback → revised version
- returning to a task in a later session
- using simple versions such as v0, v1, and v2 to make progress visible

Step 5 — Enable peer learning

Support participants to learn from each other:

- use peer review
- encourage explanation of approaches
- make feedback respectful and constructive
- ensure participation remains inclusive

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Decision point

How much opportunity will learners have to practise, receive feedback, and improve?

- Limited practice with minimal feedback
- Practice with some feedback but little revision
- Structured practice with feedback and multiple opportunities to improve

Your choice influences how well learning translates into real-world capability.

3.8.4 Example

- **Context:** A team is designing training for researchers to analyse and communicate climate data.
- **Decision:** Should participants complete a single activity, or have opportunities to improve their work?
- **Action:** They design a sequence where participants complete an initial analysis, receive peer and facilitator feedback, and then revise their work. They label the versions v0, v1, and v2 so progress is visible. They also include a follow-up discussion on what changed and why.
- **Outcome:** Participants tend to produce stronger analyses, better understand their mistakes, and gain confidence in applying the method independently.

3.8.5 In practice

👉 Use Template 9: Practice and Feedback Plan

Include:

- **what to do:** Plan how learners will receive feedback and where revision will happen
- **expected output:** A structured plan for practice and feedback during learning
- **approximate time:** 20–30 minutes

👉 Revisit Template 8: Learning Activity Design

Include:

- **what to do:** Refine one activity to include practice, feedback, and at least one revision cycle
- **expected output:** An improved activity design aligned with learning processes
- **approximate time:** 10–15 minutes

3.8.6 Key takeaways

🔥 **Key takeaway**

Learning becomes usable through realistic practice, not only exposure.

🔥 **Key takeaway**

Feedback and revision should be built into the design, not added only at the end.

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3.8.7 Before you move on

You should now have:

- designed opportunities for meaningful practice
- a plan for feedback during learning
- at least one activity that includes revision or iteration

3.8.8 Further reading (optional)

- Hattie, J., & Timperley, H. (2007) — *The Power of Feedback*
  → Supports: effective feedback that is specific, timely, and actionable
  → Why it matters: demonstrates how feedback significantly improves learning when it guides next steps
  → Source: https://doi.org/10.3102/003465430298487
- Ericsson, K. A. (2006) — *The Influence of Experience and Deliberate Practice on the Development of Superior Expert Performance*
  → Supports: meaningful, outcome-aligned practice and skill development
  → Why it matters: explains how structured practice leads to improved performance and expertise
  → Source: https://doi.org/10.1017/CBO9780511816796.003
- Nicol, D., & Macfarlane-Dick, D. (2006) — *Formative Assessment and Self-Regulated Learning*
  → Supports: integrating feedback, self-regulation, and iterative improvement
  → Why it matters: provides principles for designing feedback that supports learner improvement
  → Source: https://doi.org/10.1080/03075070600572090

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3.9 8. Assessment and Evaluation for Learning and Impact

🎯 Learning Outcomes	📍 Guiding Questions
• Distinguish between assessment (learning-focused) and evaluation (impact-focused) • Design formative and summative assessments aligned with learning outcomes • Define clear, observable indicators of success • Select appropriate and feasible methods for collecting evidence of learning and impact • Evaluate evidence to distinguish meaningful signal from noise	• How will you know if learning has occurred? • What is the difference between assessing learning and evaluating impact in this context? • Are your assessments aligned with your intended outcomes? • What indicators best reflect meaningful success? • What evidence will you collect, and how will it help you improve your training?

3.9.1 Why this matters

In the previous section, you designed opportunities for practice, feedback, and iteration within your learning activities.

This section focuses on how you recognise learning and understand impact as part of your overall training design.

Assessment should not be something added only at the end. It is part of how you design outcomes, activities, and feedback so that learning is visible.

Training is often evaluated using attendance or satisfaction. These can be useful, but they do not fully tell you:

- whether learning happened
- whether participants can apply what they learned
- whether the training made a meaningful difference

Effective training uses assessment to make learning visible and evaluation to better understand impact over time.

This section helps you move from...

"Did the training run?"
to:
"How do I know learning happened, and what changed as a result?"

3.9.2 Core concepts

Assessment and evaluation

Assessment focuses on what learners can do, while evaluation focuses on whether the training contributed to the intended outcomes or broader change.

Assessment
Assessment focuses on what learners know, produce, decide, or do. It should align with the outcomes you defined earlier.

Evaluation
Evaluation looks beyond the learning activity itself to understand change over time. It asks whether the training led to useful change in practice, confidence, behaviour, or other intended outcomes.

Indicators, evidence, and signal
Indicators are observable signs of learning or change. Evidence is what you collect to judge those indicators. Useful evidence provides signal, not just noise.

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3.9.3 Practical guidance

Step 1 — Define what success looks like

Start with your outcomes and ask:

- What would learners be able to do if learning is successful?
- What would meaningful change look like afterwards?

Define two or three clear, observable indicators.

Step 2 — Design assessment aligned with outcomes and activities

For each outcome, decide how learners will demonstrate it.

Examples might include:

- completing a task
- producing an output
- making and justifying a decision
- applying a method in context

Step 3 — Use formative and summative assessment

Include:

- formative assessment during learning to support improvement
- summative assessment at the end to judge achievement

Use formative approaches wherever possible.

Step 4 — Collect useful and feasible evidence

Use methods such as:

- observation
- outputs or artefacts
- short reflections
- brief follow-up questions
- peer or facilitator review

Choose methods you can realistically apply in your context.

Step 5 — Focus on signal, not noise

Useful signal might include learners successfully applying a method or producing a better-quality output. Noise might include attendance or positive comments alone.

Step 6 — Use evaluation to improve

Use what you learn to:

- refine activities
- identify gaps
- improve future delivery
- decide what to keep, change, or remove

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Decision point

How much effort will you invest in meaningful assessment and evaluation?

- Basic attendance and satisfaction only
- Some assessment of learning
- Clear indicators, aligned assessment, and useful evidence

Your choice affects how well you can understand learning and improve the training.

3.9.4 Example

- **Context:** A team has run a climate data training session.
- **Decision:** Should they judge success by attendance and participant satisfaction alone?
- **Action:** They define indicators linked to the outcomes, such as whether participants can interpret a dataset and explain two practical implications. They assess this through a task, observe how learners perform, and ask a short follow-up question about application after the session.
- **Outcome:** They get clearer evidence of learning and a better basis for improving the training.

3.9.5 In practice

👉 Use Template 10: Assessment & Evaluation Plan

Include:

- **what to do:** Define indicators, assessment methods, and evidence sources for one training sequence
- **expected output:** A plan for assessment and evaluation
- **approximate time:** 20–30 minutes

👉 Revisit Template 6: Alignment Table

Include:

- **what to do:** Check that each assessment method reflects the intended outcome
- **expected output:** A refined alignment between outcomes and assessment
- **approximate time:** 10–15 minutes

3.9.6 Key takeaways

🔥 **Key takeaway**

Assessment should reflect what learners need to do, not only what they have been told.

🔥 **Key takeaway**

Evaluation is most useful when it helps you distinguish meaningful signal from easy but weak indicators.

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3.9.7 Before you move on

You should now have:

- a small set of clear indicators of success
- an initial plan for aligned assessment
- a practical approach to collecting evidence of learning and impact

---

Bridging to Part 2

You have now designed a complete training experience, including outcomes, activities, practice, feedback, and assessment.

In the next part, you will turn this design into concrete training materials — including instructions, content, and resources — and learn how to make them reusable, adaptable, and shareable beyond a single delivery.

3.9.8 Further reading (optional)

- Black, P., & Wiliam, D. (1998) — *Assessment and Classroom Learning*
  → Supports: formative assessment aligned with learning outcomes
  → Why it matters: shows how assessment for learning improves understanding and performance
  → Source: https://doi.org/10.1080/0969595980050102
- Patton, M. Q. (2011) — *Developmental Evaluation: Applying Complexity Concepts to Enhance Innovation and Use*
  → Supports: evaluation focused on real-world impact and learning
  → Why it matters: introduces approaches for evaluating complex interventions like training in dynamic contexts
  → Source: https://doi.org/10.4135/9781442208636
- Guskey, T. (2000) — *Evaluating Professional Development*
  → Supports: distinguishing between different levels of training outcomes and impact
  → Why it matters: provides a structured approach to evaluating whether training leads to meaningful change
  → Source: https://doi.org/10.4135/9781452231389

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4. Part 2 - From Training Design to OER

4.1 PART 2: From Training Design to Open Educational Resources

4.1.1 1. Designing for Openness

4.1.2 2. Finding and Evaluating Existing OERs

4.1.3 3. What Makes Resources Open?

4.1.4 4. Adapting, Remixing, and Localising OERs

4.1.5 5. Sharing and Publishing OERs

4.1.6 6. Sustaining and Improving OERs

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4.2 1. Designing for Openness

4.2.1 1. Designing for Openness

🎯 Learning Outcomes	📍 Guiding Questions
• Explain how designing for openness supports reuse, adaptation, and impact • Decide when to reuse, adapt, or create resources • Identify key features of reusable and adaptable training materials • Recognise hidden assumptions that limit reuse and accessibility	• Can your training be used without you being present? • Where could existing resources be reused or adapted? • What makes a training resource easy to reuse or modify? • What assumptions are embedded in your materials?

4.2.2 Why this matters

In Part 1, you designed your training as a system — understanding your learners and context, defining outcomes, and shaping activities, practice, and feedback.

Now you start working with the materials that will bring that design to life.

You do not need to start from scratch. Effective training is often built through a combination of:

- reusing existing materials
- adapting materials to your context
- creating new content where needed

This section focuses on how to make those decisions in a practical way.

It is not about getting everything right upfront. It is about making small, deliberate choices that make your materials easier to use, adapt, and improve over time.

> **This section helps you move from...**
>
> *“Creating everything yourself for a single delivery”*
> to:
> *“Using, adapting, and creating materials that can improve and be reused over time.”*

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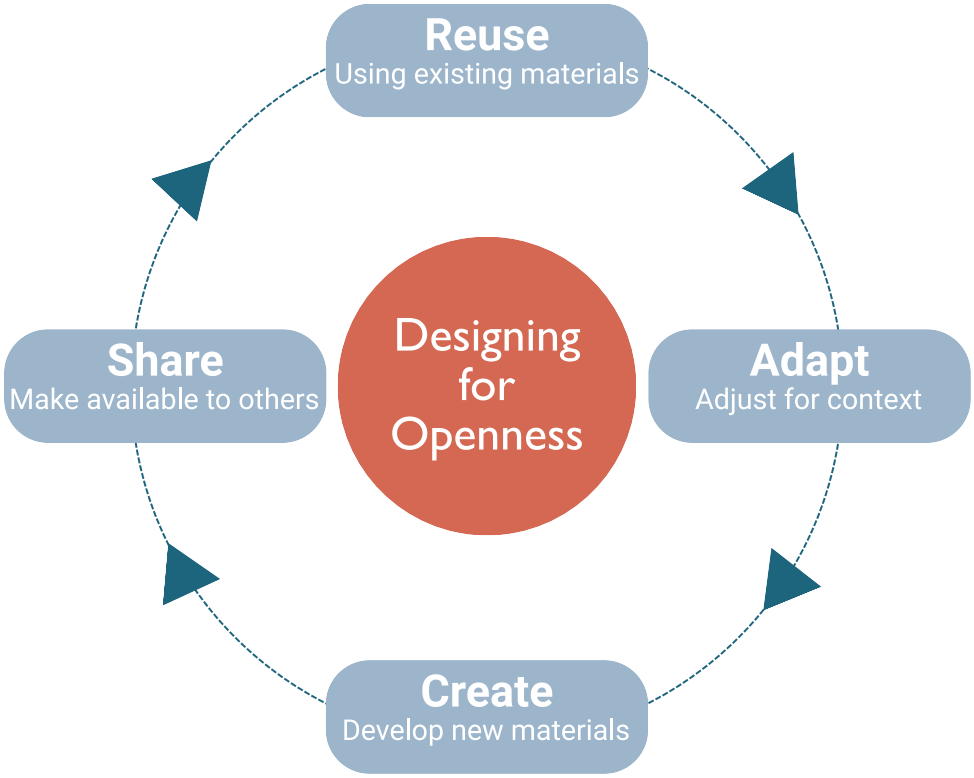

Contributing to a shared ecosystem of reusable knowledge

4.2.3 Core concepts

Openness as a design choice

Openness is a set of practical design decisions that make materials easier to use, adapt, and share over time.

Openness is not something you add at the end. It is reflected in how you structure materials, write instructions, and make assumptions visible so others can understand and use them.

Reuse, adapt, create

Training materials can be reused as they are, adapted for a different context, or created from scratch.

Most effective training combines all three. You do not need to commit to one approach — you can choose what makes sense for each part of your training.

Reusable materials

Reusable materials are more likely to be understood and used without relying heavily on your explanation.

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They do not need to be perfect or complete. They need to be clear enough that someone else (or your future self) can use them with reasonable effort.

Hidden assumptions

Materials often include unstated expectations about prior knowledge, tools, or context.

These assumptions are often linked to what you learned in Part 1 about your learners and context. Making them visible makes it easier for others to understand, adapt, and use your materials.

4.2.4 Practical guidance

Step 1 — Start with what you already have

Choose one part of your training (an activity, slides, or notes).

Ask:

- What already works well?
- What would need explanation if someone else used this?
- What feels unclear or incomplete?

You do not need to review everything. Start with one example.

Step 2 — Decide how to approach it

For that material, decide how you want to work with it:

- reuse → use it as it is
- adapt → change parts of it
- create → build something new

What approach is most practical for this part of your training?

- Reuse → when it already fits your needs
- Adapt → when it is useful but needs changes
- Create → when nothing suitable exists or adaptation would take more effort

You can make different choices for different parts of your training.

Step 3 — Improve one element

Make one small, practical improvement:

- clarify instructions
- make the activity self-contained
- add missing context (who it is for, what it does)

Focus on what would make this easier for someone else to use — not on making it perfect.

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Step 4 — Make one assumption visible

Identify one hidden assumption, for example:

- prior knowledge
- required tools or access
- context-specific examples

Make it explicit in your material so others can understand and adapt it more easily.

Step 5 — Consider future use

Ask:

- Would this still make sense if I used it again in a few months or in a different context?
- What would another facilitator need to run this with minimal explanation?

You do not need full answers yet — this helps you start noticing what supports reuse and improvement over time.

4.2.5 Example

- **Context:** A trainer has a set of workshop slides they have used multiple times.
- **Decision:** Should they reuse the slides as they are or make changes?
- **Action:** They review one activity and realise the instructions rely on what they usually say out loud. They add a short written explanation and clarify the expected output.
- **Outcome:** The activity becomes easier to run again and could be used by someone else with less explanation.

4.2.6 In practice

👉 Use Template 12: OER Workflow

Focus on:

- starting a first pass through your material
- making initial reuse, adaptation, and creation decisions

Include:

- **what to do:** Review one part of your training and decide what to reuse, adapt, create, or clarify
- **focus sections:** 0 (Context), 2 (Design Decisions)
- **expected output:** A first set of documented decisions about how you will work with this material
- **approximate time:** 20–30 minutes

👉 Revisit Template 12: OER Workflow

Include:

- **what to do:** Refine your decisions after improving one element of your material
- **expected output:** An updated set of decisions that reflect what you learned through small changes
- **approximate time:** 10–15 minutes

👉 Revisit Template 8: Learning Activity Design

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Include:

- **what to do:** Improve one activity so it can be used with minimal additional explanation
- **expected output:** A clearer, more reusable, and self-contained activity
- **approximate time:** 10–15 minutes

Suggested process

Step 1 — Select one example
Choose a small, manageable part of your training.

Step 2 — Make initial decisions
Decide whether to reuse, adapt, or create.

Step 3 — Improve clarity
Make one targeted improvement.

Step 4 — Update your decisions
Capture what changed and why in Template 12.

4.2.7 Key takeaways

Key takeaway

You do not need to redesign everything — small, focused improvements can make your materials easier to reuse and adapt.

Key takeaway

Designing for openness starts with clarity. As your materials become clearer, they also become easier to improve and reuse over time.

4.2.8 Before you move on

At this point, you should have:

- at least one example of a material you have reviewed
- an initial decision to reuse, adapt, or create
- one small improvement that makes a material easier to use without you
- one assumption made visible

These are starting points. You will revisit and refine them as you work through the rest of this module.

4.2.9 Further reading (optional)

- Wiley, D. (2014) — *The Access Compromise and the 5th R*
  → Supports: reuse and adaptation in open education
  → Why it matters: explains how openness enables meaningful reuse
  → Source: https://opencontent.org/blog/archives/3221
- UNESCO (2019) — *Recommendation on Open Educational Resources*
  → Supports: openness and reuse
  → Why it matters: defines principles for making materials accessible and adaptable
  → Source: https://www.unesco.org/en/legal-affairs/recommendation-open-educational-resources-oer

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- Hodgkinson-Williams, C., & Trotter, H. (2018) — *A social justice framework for understanding OER adoption*
  → Supports: context and accessibility
  → Why it matters: highlights how context shapes reuse and adaptation
  → Source: https://www.roarmap.org/wp-content/uploads/2018/04/hodgkinson-williams_trotter_oer_adoption.pdf

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4.3 2. Finding and Evaluating Existing OER

4.3.1 Lesson 2 — Finding & Evaluating Open Educational Resources

🎯 Learning Outcomes	📍 Guiding Questions
• Identify and locate relevant Open Educational Resources (OER) for your training • Evaluate OER for alignment, clarity, context fit, adaptability, and licensing • Recognise risks and limitations when reusing existing materials • Make explicit decisions to reuse, adapt, or create resources • Select materials that support your learners, context, and outcomes	• Where can you find relevant OER? • How do you determine whether a resource will work in your training? • What makes a resource reusable in practice? • What risks are introduced when reusing materials? • When should you reuse, adapt, or create a resource?

4.3.2 Why this matters

In the previous section, you started deciding where you might reuse, adapt, or create materials.

Now you focus on **finding and selecting resources that can support those decisions**.

There are many Open Educational Resources (OER) available. The challenge is not just finding materials — it is finding materials that will work for **your learners, your context, and your outcomes**.

Even strong materials may:

- assume different levels of knowledge
- rely on tools or access your learners do not have
- require explanation to make sense
- not align with what you want learners to achieve

This does not mean you should avoid using them. It means you need to make **practical decisions about how to use them**.

> **This section helps you move from...**
>
> *"What materials can I find?"*
> to:
> *"Which materials are worth using, adapting, or replacing?"*

4.3.3 Core concepts

> **Fit for purpose**
>
> A material is useful when it supports your learning outcomes and works for your learners and context.

A well-designed resource may still not fit your needs. Fit depends on alignment, context, and whether it can realistically be used in your setting.

> **Reusability**
>
> Reusable materials are more likely to be understood and used with limited additional explanation.

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A resource may fit your context but still require significant explanation. Improving reusability reduces that effort — for you and for others.

Contextual assumptions

All materials are created with assumptions about learners, tools, and environments.

These assumptions are often linked to what you identified in Part 1 about your learners and context. Making them visible helps you decide whether a resource will work or what needs to change.

Licensing enables action

Open licenses define what you are allowed to do with a resource.

You may identify useful materials, but what you can do with them depends on their license. You will explore this in more detail in the next lesson.

4.3.4 Practical guidance

Step 1 — Find a small set of candidates

Search for materials using:

- open repositories and institutional collections
- platforms like GitHub and Zenodo
- training materials from organisations and universities
- targeted web searches filtered for open licenses

Use specific terms that include:

- topic
- learner level
- type of material

You do not need to find the “perfect” material — aim for 1–3 promising options to work with.

Step 2 — Evaluate for fit

Use a small set of lenses to understand whether a material could work in your training:

Alignment - Does this support your learning outcomes?
- Is the level appropriate?

Clarity - Can learners follow this without heavy explanation?
- Are instructions clear enough to use?

Context fit - Are examples meaningful for your learners?
- Does it assume tools or access your learners may not have?

Adaptability - Can you realistically modify the content and format? - Is it editable and usable?

Licensing - Are you allowed to reuse or adapt it?
- Are there restrictions you need to consider?

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You do not need to analyse everything in detail. Focus on what affects how you will use the resource.

Step 3 — Identify effort and risks

Ask:

- What might learners struggle with?
- What would need explanation or adjustment?
- What assumptions does this resource make?
- How much effort would it take to make this usable?

This helps you decide whether the resource is realistic for your context.

Step 4 — Make a practical decision

How should you use this resource?

- Reuse → it fits well and needs minimal changes
- Adapt → it is useful but needs adjustments
- Create → it does not fit or would take too much effort to adapt

There is no single correct choice. The goal is to make a decision that is **realistic for your context and time**.

4.3.5 Example

- **Context:** A trainer is designing a workshop on climate data tools for community facilitators.
- **Decision:** Should they use an existing online tutorial?
- **Action:** They review the tutorial and identify assumptions about internet access, device availability, and prior coding experience. They decide to adapt one part of the tutorial and replace the rest with a simpler, locally relevant activity that they create themselves.
- **Outcome:** The training becomes more accessible and easier to deliver, while still drawing on useful elements from existing material.

4.3.6 In practice

👉 Use Template 12: OER Workflow

Include:

- **what to do:** Identify one resource and evaluate it using alignment, clarity, context fit, adaptability, and licensing
- **focus sections:** 1 (Find and Evaluate Resources), 2 (Design Decisions)
- **expected output:** A documented decision (reuse, adapt, or create) with clear reasoning
- **approximate time:** 20–30 minutes

👉 Revisit Template 8: Learning Activity Design

Include:

- **what to do:** Check whether the selected resource supports one of your activities with minimal explanation
- **expected output:** A refined, adapted, or created activity
- **approximate time:** 10–15 minutes

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Suggested process

Step 1 — Select one resource
Choose a candidate resource relevant to your training.

Step 2 — Evaluate systematically
Apply the five criteria (alignment, clarity, context fit, adaptability, licensing).

Step 3 — Decide
Choose whether to reuse, adapt, or create.

Step 4 — Test usability
Check whether it works within your activity.

4.3.7 Key takeaways

Key takeaway

You do not need to find perfect materials — you need to make informed, practical decisions about what to use, adapt, or create.

Key takeaway

Evaluating resources helps you understand what will work in your context and what will require change.

4.3.8 Before you move on

At this point, you should have:

- at least one resource identified and evaluated
- an initial decision to reuse, adapt, or create
- a clearer sense of what fits your learners and context
- at least one activity that has been reviewed or improved

These decisions are not final. You will revisit and refine them as you adapt materials and work with licensing in the next lessons.

4.3.9 Further reading (optional)

- Hilton, J. (2016) — *Open educational resources and college textbook choices*
  → Supports: effectiveness and adoption of OER
  → Why it matters: provides evidence for the impact of OER on learning and cost
  → Source: https://www.tandfonline.com/doi/full/10.1080/00221546.2016.1178814
- Wiley, D. (2014) — *The Access Compromise and the 5th R*
  → Supports: reuse and permissions in open education
  → Why it matters: explains how openness enables meaningful adaptation
  → Source: https://opencontent.org/blog/archives/3221
- UNESCO (2019) — *Recommendation on Open Educational Resources*
  → Supports: global OER principles and policy
  → Why it matters: defines frameworks for equitable access and reuse
  → Source: https://www.unesco.org/en/legal-affairs/recommendation-open-educational-resources-oer

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4.4 3. What Makes Resources Open?

4.4.1 Lesson 3 — Openness, Licensing, and Attribution

🎯 Learning Outcomes	📍 Guiding Questions
• Explain how open licenses enable reuse, adaptation, and sharing of training materials • Distinguish between different Creative Commons license types and their implications • Make informed decisions about how to license your own materials • Apply correct and consistent attribution when reusing or adapting OER • Identify risks when combining or adapting differently licensed materials	• What does “open” allow you to do in practice? • How do different licenses affect reuse and adaptation? • What license should you choose for your materials? • How do you correctly attribute reused content? • What problems arise when combining materials with different licenses?

4.4.2 Why this matters

In the previous section, you identified and evaluated resources and made initial decisions about what to reuse, adapt, or create.

Now you need to check what is **actually possible with those resources**.

Many useful materials are available online, but access does not always mean permission. Licensing affects:

- whether you can modify a resource
- whether you can combine it with others
- whether others can reuse your materials

This may influence your earlier decisions. Some resources you selected may need to be adapted differently — or created from scratch.

You do not need to master licensing in detail. You need enough understanding to make **practical, confident choices**. This is not about limiting your choices — it helps you make better ones.

> **This section helps you move from...**
>
> *“This resource looks useful”*
> to:
> *“I understand what I can do with it — and how others can use my work”*

4.4.3 Core concepts

> **Licenses**
>
> A license defines what you are allowed to do with a resource.

By default, most materials are not openly reusable. A license makes permissions explicit — for example, whether you can reuse, modify, or share a resource.

> **Open licenses**
>
> Open licenses are designed to make reuse, adaptation, and sharing possible.

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They allow you to use and build on existing materials without needing to request permission each time.

Creative Commons elements

Creative Commons (CC) licenses use a small set of conditions to communicate permissions clearly.

The main elements are:

- **BY (Attribution)** → you must credit the creator
- **SA (ShareAlike)** → adaptations must use the same license
- **NC (NonCommercial)** → cannot be used commercially
- **ND (NoDerivatives)** → cannot be modified

These combine into common licenses such as:

- **CC BY** → most open (reuse, adapt, share with attribution)
- **CC BY-SA** → must share adaptations under same license
- **CC BY-NC** → reuse and adapt, but not for commercial use
- **CC BY-ND** → reuse allowed, but no changes

You do not need to memorise all combinations — focus on what affects how you plan to use a resource.

License compatibility

When combining materials, their licenses must not conflict.

For example:

- ND (NoDerivatives) materials cannot be adapted
- SA (ShareAlike) requires you to apply the same license
- NC (NonCommercial) limits how materials can be used

Compatibility does not need to be perfect — it needs to be clear enough that your materials can be used without legal or practical conflicts.

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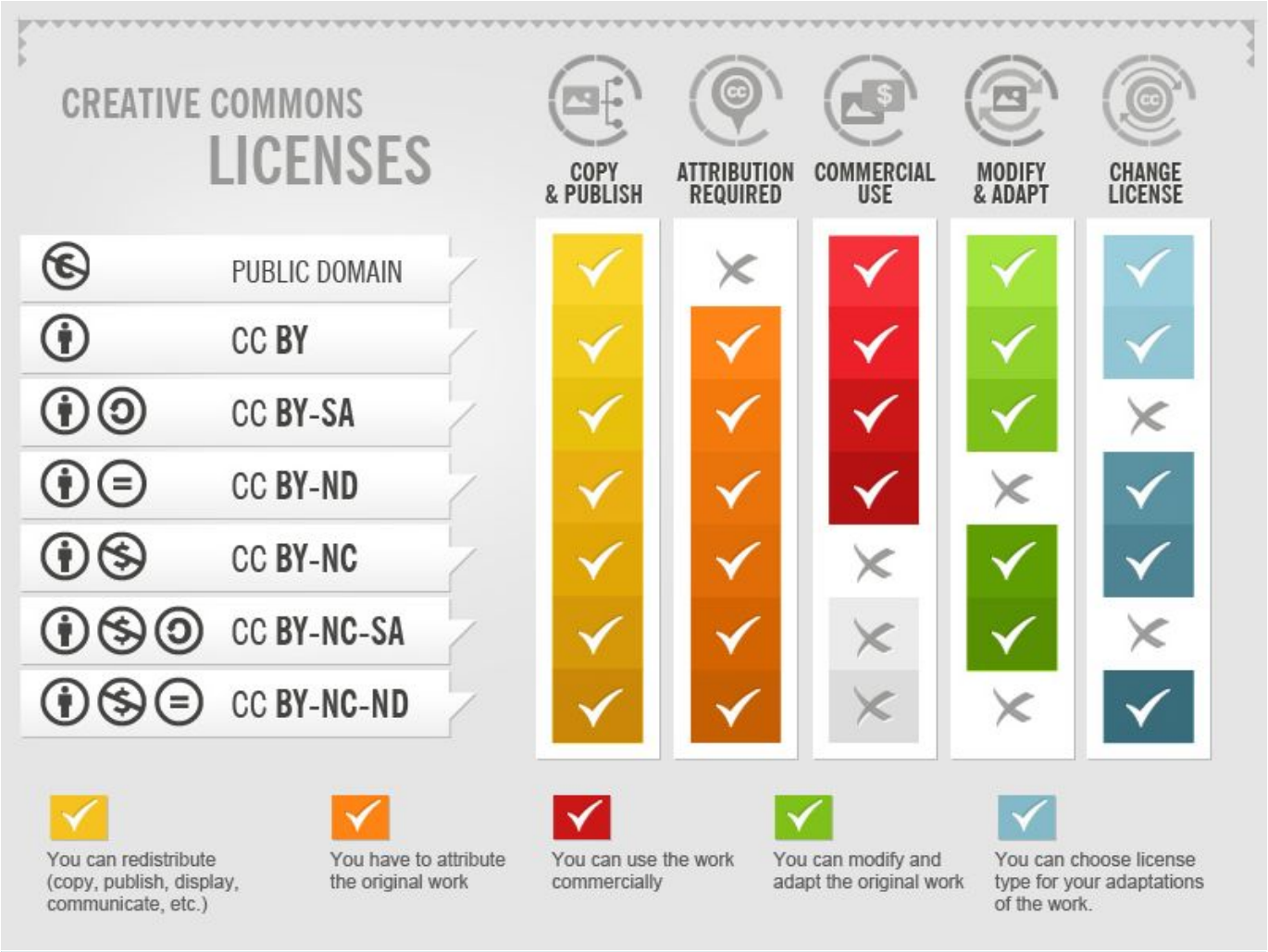

Source: Creative Commons Licenses by Foter (CC BY-SA)

Attribution

Attribution is how you credit the original creator when you reuse or adapt a resource.

A simple approach is:

- author
- title
- source
- license

Clear attribution supports reuse, transparency, and trust.

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4.4.4 Practical guidance

Step 1 — Identify the license

For each resource:

- find the license statement
- note the type (e.g. CC BY, CC BY-SA)
- check whether it is clearly stated

If no license is visible, treat the material as **not openly reusable** and consider alternatives.

Step 2 — Focus on what matters for your use

Ask a few practical questions based on how you plan to use the resource:

- Can I modify this material?
  → (ND means no)
- Can I combine it with other materials?
  → (SA may affect this)
- Can it be used in my context?
  → (NC may limit some uses)

You do not need to interpret every detail. Focus on what affects your intended use.

Step 3 — Check combinations

If you are using multiple resources:

- check whether their licenses can work together
- look for potential conflicts

What should you do if there is a conflict?

- Replace one resource with another that has compatible licensing or create one from scratch
- Simplify your design
- Keep materials separate instead of combining

These are design choices. Choose what is most practical for your context.

Step 4 — Choose a license for your materials

How open do you want your materials to be?

- More open (e.g. CC BY) → easier for others to reuse and adapt
- More controlled (e.g. NC or SA) → more conditions on reuse

Choose based on:

- your goals for reuse
- your context and stakeholders
- how you expect others to use your materials

There is no single correct choice — focus on what supports your intended use.

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Step 5 — Add attribution

When reusing or adapting:

- include author, title, source, and license
- keep attribution visible and consistent

Clear attribution helps others understand where materials come from and supports reuse and adaptation.

4.4.5 Example

- **Context:** A trainer is combining two OER resources into one workshop.
- **Decision:** Can both resources be adapted and combined?
- **Action:** They identify that one resource has a CC BY-ND license, which does not allow modification. Since adaptation is needed, they replace it with a CC BY resource.
- **Outcome:** The final material can be adapted and reused more easily, supporting future use.

4.4.6 In practice

👉 Use Template 12: OER Workflow

Focus on:

- checking what is allowed for the resources you selected
- updating your earlier reuse and adaptation decisions

Include:

- **what to do:** Identify the license for each selected resource and note what is allowed (reuse, adapt, combine)
- **focus sections:** 1 (Find and Evaluate Resources), 2 (Design Decisions)
- **expected output:** An updated set of licensing constraints and decisions for each resource
- **approximate time:** 20–30 minutes

👉 Revisit Template 12: OER Workflow

Include:

- **what to do:** Check whether selected resources can be combined without conflicts and adjust your plan if needed
- **expected output:** A set of resources that can be used together, or clear decisions about what to create or keep separate
- **approximate time:** 10–15 minutes

Suggested process

Step 1 — List your resources
Write down selected resources and their licenses.

Step 2 — Identify key conditions
Note BY, SA, NC, ND elements.

Step 3 — Review decisions
Adjust reuse/adapt/create choices if needed.

Step 4 — Check combinations
Ensure your materials can work together or revise your approach.

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4.4.7 Key takeaways

Key takeaway

You do not need to understand every detail of licensing — focus on what affects how you reuse, adapt, and combine materials.

Key takeaway

Licensing helps you make better decisions about what is possible and how your materials can be used by others.

4.4.8 Before you move on

At this point, you should have:

- a list of resources with identified licenses
- a clearer understanding of key permissions (BY, SA, NC, ND)
- updated decisions about reuse, adaptation, or creation
- a plan for including attribution in your materials

These decisions may change as you adapt your materials. The goal is to make choices that are clear enough to move forward.

4.4.9 Further reading (optional)

- Creative Commons (n.d.) — *About The Licenses*
  → Supports: understanding license types and permissions
  → Why it matters: provides authoritative guidance on how licenses work
  → Source: https://creativecommons.org/licenses/
- Wiley, D. (2014) — *The Access Compromise and the 5th R*
  → Supports: reuse and adaptation in open education
  → Why it matters: explains how openness enables meaningful reuse
  → Source: https://opencontent.org/blog/archives/3221
- UNESCO (2019) — *Recommendation on Open Educational Resources*
  → Supports: global OER policy and principles
  → Why it matters: defines frameworks for equitable access and reuse
  → Source: https://www.unesco.org/en/legal-affairs/recommendation-open-educational-resources-oer

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4.5 4. Adapting, Remixing, and Localising OER

4.5.1 Lesson 4 — Adapting and Localising Open Educational Resources

🎯 Learning Outcomes	📍 Guiding Questions
• Adapt OER to align with your learning outcomes, learners, and context	• What needs to change for this resource to work in your context?
• Identify and modify assumptions embedded in existing materials	• What assumptions are embedded in the original material?
• Distinguish between surface-level and structural adaptation	• How much adaptation is required?
• Make informed decisions about what to change, retain, or remove	• What should you keep, modify, or remove?
• Document adaptations to support reuse and transparency	• How do you make your adaptations visible to others?

4.5.2 Why this matters

In the previous section, you clarified what you are allowed to do with the resources you selected.

Based on what is possible, you now focus on **making those resources work in your training**.

Even when a resource is relevant and can be reused, it often does not fit your learners, context, or constraints without modification.

If you use materials without adapting them, you may introduce:

- examples that do not resonate with your learners
- activities that assume tools or infrastructure that are not available
- sequences that do not match your learning outcomes
- explanations that rely on prior knowledge your learners do not have

Adapting materials is a normal and expected part of working with OER. You do not need to adapt everything — even small, targeted changes can make a resource more usable.

> **This section helps you move from...**
>
> *"This resource is usable"*
> to:
> *"This resource works in my context"*

4.5.3 Core concepts

> **Adaptation**
>
> Adaptation is the process of modifying a resource so it aligns with your learners, context, and outcomes.

Adaptation is expected when working with OER. It helps ensure materials are usable and relevant in your setting.

> **Localisation**
>
> Localisation is a type of adaptation that focuses on language, examples, culture, and context.

It helps make materials more accessible and meaningful for specific groups of learners.

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Levels of adaptation

Adaptation can happen at different levels, depending on how much needs to change.

- Surface → wording, formatting, terminology
- Structural → sequence, activities, emphasis
- Contextual → goals, assumptions, delivery conditions

You do not need to change everything — choose the level of adaptation that fits your needs.

Design trade-offs

Adapting materials involves balancing what you keep from the original with what you change to fit your context.

You are not required to preserve everything. Changing or removing parts of a resource is often necessary to make it work effectively.

4.5.4 Practical guidance

Step 1 — Identify assumptions

Review the resource and ask:

- What does this assume about learners' prior knowledge?
- What tools or infrastructure are required?
- What context or examples are used?

Make implicit assumptions visible so you can work with them directly.

Step 2 — Identify what does not fit

Compare the resource against your training:

- learning outcomes
- learner realities
- constraints (time, access, language, tools)

Highlight where there are mismatches.

Step 3 — Decide what to change

For each mismatch, decide:

- keep → if it already fits
- modify → if it can be adapted
- remove → if it does not support your goals

You are making practical decisions under real constraints — not ideal ones. You do not need to change everything — focus on what will have the most impact.

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Step 4 — Apply targeted adaptations

Make small, purposeful changes such as:

- adjusting examples to local context
- simplifying or restructuring activities
- changing sequencing to match your design
- removing unnecessary complexity

Start with changes that make the biggest difference.

Step 5 — Document your changes

Record:

- what you changed
- why you changed it
- what assumptions were addressed

This helps others — and your future self — understand and reuse your work more easily.

Step 6 — Check coherence

Does the adapted resource still function as a learning experience?

- Works well → ready to use
- Works partially → refine further
- Does not yet work → revisit your changes

Coherence does not need to be perfect — it needs to be workable in your context.

4.5.5 Example

- **Context:** A trainer is adapting an OER workshop on data analysis for use in a low-resource setting.
- **Decision:** How much adaptation is required to make the workshop usable?
- **Action:** They identify assumptions about internet access, software availability, and prior technical knowledge. They replace online tools with offline alternatives, simplify exercises, and reorder activities to build foundational understanding.
- **Outcome:** The adapted workshop is more accessible and better aligned with learner needs, while still achieving the intended learning outcomes.

4.5.6 In practice

👉 Use Template 12: OER Workflow

Focus on:

- identifying what needs to change
- documenting your adaptation decisions
- refining your materials so they work in your context

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Include:

- **what to do:** Identify what to keep, modify, or remove and document your adaptation decisions
- **focus sections:** 3 (Adaptation and Localisation Log)
- **expected output:** An updated adaptation log with clear decisions and rationale
- **approximate time:** 20–30 minutes

👉 Revisit Template 8: Learning Activity Design

Include:

- **what to do:** Update one activity using your adapted material so it works with minimal additional explanation and fits your context
- **expected output:** A revised, context-appropriate, and more reusable activity
- **approximate time:** 10–15 minutes

Suggested process

Step 1 — Surface assumptions
List assumptions embedded in the original resource.

Step 2 — Identify mismatches
Compare with your learners and context.

Step 3 — Make decisions
Decide what to keep, modify, or remove.

Step 4 — Apply changes
Implement targeted adaptations.

Step 5 — Review and refine
Check whether the material works and update your decisions if needed.

4.5.7 Key takeaways

Key takeaway

Adaptation helps make materials usable and relevant in real contexts.

Key takeaway

You do not need to adapt everything — focus on changes that make the biggest difference for your learners and context.

4.5.8 Before you move on

At this point, you should have:

- a set of resources with identified assumptions
- documented decisions on what to keep, modify, or remove
- at least one adapted activity
- materials that better reflect your learners and context

These are working versions. You will continue refining them as you prepare materials for sharing and reuse.

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4.5.9 Further reading (optional)

- Wiley, D. (2014) — *The Access Compromise and the 5th R*
  → Supports: adaptation and reuse in open education
  → Why it matters: explains how openness enables modification and improvement
  → Source: https://opencontent.org/blog/archives/3221
- UNESCO (2019) — *Recommendation on Open Educational Resources*
  → Supports: localisation and equitable access
  → Why it matters: emphasises adapting resources for different contexts
  → Source: https://www.unesco.org/en/legal-affairs/recommendation-open-educational-resources-oer
- Hodgkinson-Williams, C., & Trotter, H. (2018) — *A social justice framework for understanding OER adoption*
  → Supports: contextual and equity considerations in OER
  → Why it matters: highlights how context shapes effective use of OER
  → Source: https://www.roarmap.org/wp-content/uploads/2018/04/hodgkinson-williams_trotter_oer_adoption.pdf

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4.6 5. Sharing and Publishing OER

4.6.1 Lesson 5 — Sharing and Publishing Open Educational Resources

🎯 Learning Outcomes	📍 Guiding Questions
• Prepare OER so they can be used without additional explanation	• When is a resource ready to share?
• Select appropriate platforms for sharing and publishing materials	• Where should you publish your materials?
• Ensure materials are discoverable and understandable by others	• How do you make your materials understandable to others?
• Document resources clearly to support reuse and adaptation	• What information do others need to reuse your work?
• Make decisions about when a resource is ready to share	• How do you ensure your materials can be found and used?

4.6.2 Why this matters

In the previous section, you adapted materials so they work in your context.

Once your materials work in your context, the next step is to **make them usable for others**.

For others to reuse and adapt your work, your materials need to be clear, understandable, and easy to use.

If materials are not clearly documented and shared appropriately, they:

- cannot be reused without your explanation
- are difficult to find
- are misinterpreted or misused
- fail to contribute to broader impact

Sharing is not just uploading files.

It is about making your materials **usable, understandable, and reusable by others**.

You do not need to make everything perfect before sharing. The goal is to make materials clear enough that others can use them with reasonable effort.

> **This section helps you move from...**
>
> *"These materials work for my training"*
> to:
> *"These materials can be used and adapted by others"*

4.6.3 Core concepts

> **Publishability**
>
> A resource is publishable when it is clear enough to be shared and used by others.

This builds on reusability. Materials should not rely on your explanation and should include enough context for others to understand how to use them.

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Documentation

Documentation provides the context needed to understand and use a resource.

This includes purpose, audience, instructions, and assumptions that were made during design and adaptation. Making these explicit helps others reuse and adapt your materials.

Discoverability

Discoverability refers to how easily others can find your materials and understand what they are for.

Clear titles, descriptions, and keywords help others locate your work and decide whether it is useful to them.

Platform choice

Different platforms support different goals, formats, and audiences.

This builds on reusability. Materials that are reusable can be made publishable by adding sufficient context and documentation. Where you publish affects who can find, use, and adapt your materials. Choosing a platform is a practical decision based on how your materials will be used.

4.6.4 Practical guidance

Step 1 — Check readiness

Review your materials and ask:

- Can someone use this without my explanation?
- Are instructions clear enough to follow?
- Is the purpose of the material understandable?

Materials do not need to be complete — they need to be clear enough to use.

Step 2 — Add essential documentation

Include the information others need to understand and use your materials:

- purpose and learning outcomes
- intended audience
- instructions for use
- assumptions and requirements
- licensing and attribution

Focus on what would help someone else use your material without additional explanation.

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Step 3 — Improve clarity and usability

Make small, targeted improvements:

- simplify language
- clarify instructions
- remove unnecessary complexity
- ensure consistency across materials

Focus on changes that make the biggest difference to usability.

Step 4 — Choose where to publish

Where should you share your materials?

- Versioned, technical audience → platforms like GitHub
- Citable output → repositories like Zenodo
- Broad accessibility → web pages or institutional platforms

Choose based on:

- who needs to access the materials
- how they will be used
- what formats you are sharing

There is no single correct platform — focus on what best supports your goals.

Step 5 — Make materials discoverable

Help others find and understand your materials:

- use clear and descriptive titles
- add keywords and summaries
- provide context for how the material can be used

This helps others quickly decide whether your materials are relevant and worth using.

4.6.5 Example

- **Context:** A trainer has adapted a set of workshop materials for a specific audience and wants to share them.
- **Decision:** Are the materials ready to be published and reused by others?
- **Action:** They review the materials and realise that instructions are incomplete and assumptions about prior knowledge are not stated. They add documentation, clarify steps, include learning outcomes, and select a platform where the materials can be easily accessed and cited.
- **Outcome:** The materials become understandable, reusable, and discoverable, enabling others to adapt them for different contexts.

4.6.6 In practice

👉 Use Template 12: OER Workflow

Focus on:

- preparing your materials so they can be shared and used by others
- refining documentation, context, and clarity

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Include:

- **what to do:** Ensure your materials include clear documentation, context, and licensing, and are ready to be shared
- **focus sections:** 4 (Prepare for Sharing)
- **expected output:** A set of materials prepared for sharing with sufficient documentation and clarity
- **approximate time:** 20–30 minutes

👉 Revisit Template 9: Practice & Feedback Plan

Include:

- **what to do:** Identify where someone using your materials might need clarification or support, and how feedback could improve usability
- **expected output:** A clearer understanding of how your materials perform without you and where improvements are needed
- **approximate time:** 10–15 minutes

Suggested process

Step 1 — Review for independence
Check whether materials can be used without you.

Step 2 — Add documentation
Provide missing context and instructions.

Step 3 — Improve clarity
Refine language and structure.

Step 4 — Test usability
Identify where users may struggle or need support.

Step 5 — Refine and update
Improve materials based on what you learned.

4.6.7 Key takeaways

> 🔥 **Key takeaway**
>
> Sharing is about usability — making it possible for others to understand and use your materials.

> 🔥 **Key takeaway**
>
> Materials do not need to be complete to be shared — they need to be clear enough that others can use them with reasonable effort.

4.6.8 Before you move on

At this point, you should have:

- materials that can be used without your explanation
- clear documentation (purpose, audience, instructions, assumptions)
- at least one activity that is self-contained
- a decision about where and how to share your materials

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These materials are ready to be shared, but they can still improve. You will continue refining them as they are used and as you receive feedback.

4.6.9 Further reading (optional)

- UNESCO (2019) — *Recommendation on Open Educational Resources*
  → Supports: sharing and access to OER
  → Why it matters: defines principles for making resources widely available and reusable
  → Source: https://www.unesco.org/en/legal-affairs/recommendation-open-educational-resources-oer
- Atenas, J., & Havemann, L. (2014) — *Questions of quality in repositories of open educational resources*
  → Supports: discoverability and usability of OER
  → Why it matters: highlights factors that influence whether resources are found and used
  → Source: https://www.irrodl.org/index.php/irrodl/article/view/2083
- Wiley, D., & Hilton, J. (2018) — *Defining OER-enabled pedagogy*
  → Supports: reuse and sharing practices
  → Why it matters: connects openness to meaningful educational impact
  → Source: https://opencontent.org/blog/archives/5009

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4.7 Lesson 6 — Sustaining and Improving Open Educational Resources

🎯 Learning Outcomes	📍 Guiding Questions
• Understand the lifecycle of OER beyond initial creation and sharing • Identify realistic approaches to maintaining and updating materials • Collect and use feedback to improve training materials • Make decisions about when to update, maintain, or archive resources • Ensure materials remain usable and relevant over time	• What happens to materials after they are shared? • How do you keep resources relevant over time? • What feedback should you collect and use? • What does “sustainable” mean in practice? • When should you update, maintain, or stop maintaining a resource?

4.7.1 Why this matters

In the previous section, you prepared and shared your materials so they can be used by others.

Your materials **are now being used by others**.

Once materials are shared, they are used in real contexts, by different people, under changing conditions. Over time, they may become:

- outdated
- misaligned with new contexts
- unclear or incomplete
- difficult to use without revision

Most OER do not fail because they are poorly designed.
They become less useful when they are **not maintained, improved, or reused effectively**.

This section focuses on how to make practical decisions about maintaining and improving your materials over time.

> **This section helps you move from...**
>
> *“My materials are shared”*
> to:
> *“My materials remain usable and improve over time”*

4.7.2 Core concepts

> **OER lifecycle**
>
> OER exist within a cycle of use, feedback, and improvement.

Reuse → Adapt → Share → Use → Gather feedback → Improve → (repeat)

Materials gain value through use and iteration, not just publication.

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Sustainability

Sustainability is about making realistic decisions about how materials are maintained over time.

Not all resources need continuous updates. Sustainability depends on how materials are used, how relevant they remain, and what capacity is available.

Feedback for improvement

Feedback shows how materials perform in real contexts.

This builds on earlier work where you designed practice and feedback. Here, feedback helps identify what needs to improve in the materials themselves.

Versioning and transparency

Versioning tracks changes over time and helps others understand what has changed and why.

Clear versioning supports reuse, trust, and ongoing improvement.

4.7.3 Practical guidance

Step 1 — Identify how materials will be used

Clarify:

- who is using the materials
- in what contexts
- how frequently they are used

This helps you decide what level of maintenance is realistic for you.

Step 2 — Collect meaningful feedback

Focus on feedback that helps you improve the material:

- where learners struggle or disengage
- what facilitators need to explain or adjust
- what does not work in different contexts

Avoid collecting feedback that does not lead to action.

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Step 3 — Decide how to respond

What level of maintenance is appropriate?

- Active maintenance → regularly update and improve
- Occasional updates → revise when needed
- No maintenance → archive and do not update

Choose based on:

- relevance
- usage
- available time and capacity

There is no single correct approach — choose what is realistic for you.

Step 4 — Make targeted improvements

Prioritise changes that:

- improve clarity and usability
- address recurring issues
- adapt to new contexts

Focus on changes that make the biggest difference rather than trying to improve everything.

Step 5 — Track and communicate changes

Record:

- what changed
- why it changed
- which version is current

This helps others understand how to use your materials and what to expect.

Step 6 — Re-evaluate fit

Do these materials still serve their intended purpose?

- Yes → continue using and maintaining
- Partially → adapt further
- No → retire or replace

All of these are valid outcomes. This step helps you decide what to do next.

4.7.4 Example

- **Context:** A trainer has shared a set of workshop materials that are now being used by others in different regions.
- **Decision:** Should the materials be actively maintained or left as they are?
- **Action:** They collect feedback from facilitators and identify recurring issues with unclear instructions and context-specific examples. They prioritise updates to improve clarity and generalise examples, and release a revised version with documented changes.

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- **Outcome:** The materials become easier to use across contexts, leading to broader adoption and more effective reuse.

4.7.5 In practice

👉 Use Template 12: OER Workflow

Focus on:

- defining how your materials will be used and maintained
- planning how feedback will inform improvements
- making realistic decisions about sustainability

Include:

- **what to do:** Define how your material will be used, what feedback you will collect, and what level of maintenance is realistic
- **focus sections:** 5 (Sustainability and Improvement)
- **expected output:** A sustainability plan that includes feedback approach, maintenance level, and update strategy
- **approximate time:** 20–30 minutes

👉 Revisit Template 12: OER Workflow

Focus on:

Versioning

Include:

- **what to do:** Define how you will track changes and what counts as a new version
- **expected output:** A simple versioning approach that makes updates visible and understandable
- **approximate time:** 10–15 minutes

Suggested process

Step 1 — Identify use patterns
Clarify who will use the material and how often.

Step 2 — Define feedback sources
Decide what feedback is meaningful and how it will be collected.

Step 3 — Choose maintenance level
Select active, occasional, or no maintenance.

Step 4 — Plan updates
Identify what will realistically be improved over time.

Step 5 — Define versioning
Decide how changes will be tracked and communicated.

Step 6 — Repeat as needed
Update your plan as your materials are used and improved.

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4.7.6 Key takeaways

Key takeaway

OER gain value through use and improvement, not just publication.

Key takeaway

Sustainable materials are those that are maintained in ways that are realistic for your context and capacity.

4.7.7 Before you move on

At this point, you should have:

- a clearer understanding of how your materials will be used
- a plan for collecting meaningful feedback
- a defined maintenance approach (active, occasional, or none)
- a simple plan for tracking and communicating updates

These are working plans. You can adjust them as your materials are used and as your context changes.

4.7.8 Further reading (optional)

- UNESCO (2019) — *Recommendation on Open Educational Resources*
  → Supports: sustainability and reuse of OER
  → Why it matters: emphasises long-term access, adaptation, and collaboration
  → Source: https://www.unesco.org/en/legal-affairs/recommendation-open-educational-resources-oer
- Wiley, D., & Hilton, J. (2018) — *Defining OER-enabled pedagogy*
  → Supports: reuse and iterative improvement
  → Why it matters: links openness to continuous improvement and impact
  → Source: https://opencontent.org/blog/archives/5009
- Atenas, J., Havemann, L., & Timmermann, C. (2020) — *Critical literacies for open educational practices*
  → Supports: reflective and adaptive use of OER
  → Why it matters: highlights the importance of context and iteration in OER use
  → Source: https://doi.org/10.5334/jime.576

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5. Conclusion

This workbook has guided you through a complete design journey:

- from **understanding training as an intervention in a system**
- to **designing learning experiences grounded in learners and context**
- to **aligning outcomes, activities, and assessment**
- to **building in practice, feedback, and iteration**
- to **preparing materials for reuse, adaptation, and sharing as OERs**
- to **sustaining and improving those materials over time**

These elements are not separate. They are interconnected and iterative.

Effective training design links:

- **intentional design decisions**
- **meaningful learning experiences**
- **practical application in real contexts**
- **materials that can be reused, adapted, and improved**
- **ongoing learning through feedback and iteration**

5.0.1 Key messages to carry forward

- **Training is an intervention**
  Its impact depends on systems, context, and what learners can actually do afterwards.
- **Learners and context matter**
  Good design starts with understanding realities, not assumptions.
- **Learning requires participation**
  Practice, feedback, and iteration turn knowledge into capability.
- **Materials extend your impact**
  What you create can be reused, adapted, and improved by others—and by your future self.
- **Impact goes beyond your participants**
  Your training influences how others teach, learn, and act in their own contexts.
- **Improvement is continuous**
  Training and materials are not finished after delivery—they evolve through use, feedback, and adaptation.

5.0.2 Your role going forward

You now have the tools to design training that is:

- realistic
- usable
- adaptable
- and capable of contributing to meaningful change over time

What matters next is action.

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Tip

Start where you are. You do not need to redesign everything at once.

You might:

- apply one section of this workbook to an existing training
- improve a single activity using clearer structure and feedback
- revisit your outcomes and alignment
- adapt one resource instead of creating from scratch
- prepare one set of materials for reuse or sharing
- improve materials based on feedback from real use

5.0.3 Next steps

- **Apply the workbook in your own context**
  Use the templates and guidance to design or refine your training.
- **Revisit and improve**
  Return to your materials after delivery and refine them based on what you learn.
- **Share and contribute**
  Make your work available for others to use, adapt, and build on.
- **Stay engaged with communities**
  Learning, design, and reuse are strengthened through collaboration.

Note

Effective training is not something you deliver once.
It is something you design, use, learn from, and improve—over time, and with others.

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6. Activities

6.1 Activity Templates

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7. Resources

7.1 Glossary

This glossary defines key concepts used throughout the workbook.
Terms are grouped to reflect how they are used in practice.

7.1.1 Systems and Impact

Training as an intervention
Training is a deliberate action within a broader system of people, resources, relationships, constraints, and external influences. Its effectiveness depends on how it interacts with this system.

Theory of Change
A structured explanation of how training activities are expected to lead to outputs, outcomes, and longer-term impact.

Outputs, outcomes, and impact
- **Outputs**: Immediate products of training (e.g. completed tasks, materials)
- **Outcomes**: What participants do differently after training
- **Impact**: Longer-term change resulting from those outcomes

Constraints (system conditions)
Real-world factors such as time, access, infrastructure, and power dynamics that shape what is possible in training design. Constraints are not just limitations—they are conditions you design within.

Positionality
The role, background, and influence of the facilitator or designer, which shapes decisions, priorities, and participation.

7.1.2 Learners and Context

Primary and secondary audiences
- **Primary audience**: Direct participants in the training
- **Secondary audience**: People those participants later support, train, or influence

Needs assessment
A practical process to understand what learners need to do, what they already know, and what constraints they face.

Learner realities
The conditions shaping participation and learning, including access, time, language, culture, and responsibilities.

Lived experience and prior knowledge
The knowledge, mental models, and experiences learners bring, which new learning should connect to.

7.1.3 Learning and How It Works

Cognitive load
The amount of information learners can process at once; excessive load reduces understanding and retention.

Retrieval
Recalling knowledge from memory, which strengthens learning more effectively than re-exposure.

Reinforcement (spacing)
Revisiting ideas over time and across contexts to improve retention and transfer.

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Social learning
Learning through interaction, discussion, collaboration, and shared problem-solving.

Active learning
Learning through doing, discussing, analysing, or applying, rather than passively receiving content.

Scaffolding
Providing structured support early in learning and gradually reducing it as learners gain independence.

Psychological safety
A learning environment where participants feel safe to contribute, make mistakes, and revise their thinking.

7.1.4 Design and Alignment

Learning outcomes
Clear, observable statements describing what learners should be able to do by the end of the training.

Backward design
A design approach that starts with learning outcomes, defines evidence of success, then designs activities and supporting content.

Alignment (constructive alignment)
Ensuring learning outcomes, activities, and assessment support the same capabilities.

Activities (learning activities)
Structured tasks where learners practise, apply, and demonstrate learning.

Content (supporting content)
Information and resources that support participation in activities, not the driver of design.

Misalignment
A disconnect between outcomes, activities, assessment, or content that reduces effectiveness.

Fit for purpose
The extent to which a resource or activity supports the intended learning outcomes and works in the specific learner and context conditions.

7.1.5 Participation, Co-Design, and Co-Creation

Co-design
Participants help shape decisions about the training (e.g. priorities, structure, activities).

Co-creation
Participants contribute ideas, examples, or outputs during activities or material development.

Meaningful participation
Learners actively contribute and have influence, rather than passively receiving content.

Trade-offs
Design decisions balancing competing priorities such as time, depth, participation, and complexity.

7.1.6 Practice, Feedback, and Iteration

Practice
Opportunities for learners to apply knowledge or skills in realistic tasks aligned with outcomes.

Feedback
Information that helps learners improve; effective feedback is specific, timely, and actionable.

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Iteration (learning)
Cycles of attempt, feedback, and revision that support improvement in learner performance.

Iteration (design and materials)
Ongoing cycles of use, feedback, and improvement that refine training design and materials over time.

Peer learning
Learning through interaction, feedback, and collaboration between participants.

7.1.7 Assessment and Evaluation

Assessment
Methods used to determine what learners can do, produce, or decide, aligned with learning outcomes.

Evaluation
Judging whether the training contributed to intended outcomes or broader change over time.

Indicators
Observable signs that learning or change has occurred.

Evidence
Information collected to assess indicators and judge success.

Signal vs noise
Distinguishing meaningful evidence of learning or impact (signal) from less useful or misleading data (noise).

7.1.8 Open Educational Resources (OER) and Materials

Open Educational Resources (OER)
Teaching and learning materials that are openly licensed, allowing reuse, adaptation, and sharing.

Training materials
Resources used to support training (e.g. slides, activities, guides). These can be reused, adapted, and shared independently of a specific training delivery.

Reuse, adapt, create
A practical decision framework for working with materials:
- **Reuse**: use as is
- **Adapt**: modify to fit context
- **Create**: develop new materials when needed

Reusability
The extent to which a material can be understood and used without additional explanation.

Adaptation
Modifying a resource so it aligns with learners, context, and outcomes.

Localisation
Adapting materials to reflect language, culture, and local context.

Publishability
The extent to which a resource is clear, documented, and ready to be shared and used by others.

Documentation
Information that explains how and why a resource should be used, including purpose, audience, and instructions.

Discoverability
How easily others can find and understand a resource.

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Versioning
Tracking changes to materials over time, including what changed, why, and which version is current.

OER lifecycle
The ongoing process through which materials are reused, adapted, shared, used, improved, and reused again.

Sustainability (of materials)
Making realistic decisions about how materials are maintained, updated, or archived based on use, relevance, and available capacity.

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7.2 Bloom's Taxonomy — Quick Guide for Writing Learning Outcomes

Bloom's taxonomy helps you choose **action verbs** that describe what learners should be able to *do*.

Use it as a guide — not a checklist.

How to use this

- Start with what learners need to **do in practice**
- Then choose a verb that reflects that action
- Avoid vague verbs like **understand**, **know**, or **learn** unless you make them more specific

7.2.1 Bloom's levels and example verbs

1. Remember

Use when learners need to recall information, terms, or basic facts.

Example verbs:

- define
- list
- identify
- name
- recall
- recognise

Example outcome:

Identify the main components of a system map.

2. Understand

Use when learners need to explain ideas, describe meaning, or make sense of concepts.

Example verbs:

- explain
- summarise
- interpret
- describe
- classify
- compare

Example outcome:

Explain the difference between outputs, outcomes, and impact.

3. Apply

Use when learners need to use knowledge, methods, or tools in a practical task.

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Example verbs:

- use
- apply
- demonstrate
- implement
- complete
- carry out

Example outcome:

Apply backward design to structure a short training session.

4. Analyse

Use when learners need to examine relationships, identify patterns, or diagnose problems.

Example verbs:

- analyse
- distinguish
- examine
- diagnose
- differentiate
- investigate

Example outcome:

Analyse a training design to identify points of misalignment.

5. Evaluate

Use when learners need to make judgements based on criteria.

Example verbs:

- evaluate
- assess
- justify
- critique
- prioritise
- judge

Example outcome:

Evaluate whether a training activity provides meaningful evidence of learning.

6. Create

Use when learners need to produce, design, or adapt something new.

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Example verbs:

- design
- create
- develop
- construct
- adapt
- produce

Example outcome:

Design a short training activity aligned with a learning outcome.

Important: You do NOT need to move through levels sequentially

> **Common misconception**
>
> Training does not need to progress step-by-step from *remember → understand → apply → analyse → evaluate → create*.
>
> Choose the level that matches what learners actually need to do.

- You can design directly for **apply**, **analyse**, or **create**
- Not all training needs all levels
- Lower levels are not always a prerequisite for higher levels

For example:

- If learners need to **use a tool**, start at **apply**
- If learners need to **diagnose issues**, start at **analyse**
- If learners need to **produce outputs**, start at **create**

7.2.2 Practical guidance

> **Important**
>
> Higher Bloom's levels are not automatically better.
>
> Choose the level that matches what learners actually need to do in their real context.

For example:

- If learners need to **follow a process correctly**, **apply** may be enough
- If learners need to **diagnose problems**, **analyse** may be more appropriate
- If learners need to **develop their own materials or plans**, **create** may be appropriate

7.2.3 Avoid vague learning outcomes

Avoid outcomes such as:

- Understand climate data
- Know the principles of good training
- Learn about alignment

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These do not make performance visible.

Make them more specific

Instead of:

- **Understand climate data**

Use:

- **Interpret a local climate dataset to identify practical risks**

Instead of:

- **Learn about alignment**

Use:

- **Identify and correct misalignment between learning outcomes, activities, and assessment**

7.2.4 Quick check for a strong learning outcome

A strong learning outcome is:

- observable
- action-oriented
- relevant to practice
- possible to assess

7.2.5 Reminder

Key principle

Choose verbs that match real-world performance.

The goal is not to sound academic. The goal is to make learning visible.

7.2.6 Further reading

- Bloom, B. S., Engelhart, M. D., Furst, E. J., Hill, W. H., & Krathwohl, D. R. (1956) — *Taxonomy of Educational Objectives: The Classification of Educational Goals. Handbook I: Cognitive Domain*
  → Supports: original classification of learning objectives into levels of cognitive complexity
  → Why it matters: establishes the foundational framework for defining and structuring learning outcomes
  → Source: https://eclass.uoa.gr/modules/document/file.php/PPP242/Benjamin%20S.%20Bloom%20-%20Taxonomy%20of%20Educational%20Objectives%2C%20Handbook%201_%20Cognitive%20Domain-Addison%20Wesley%20Publishing%20Company%20%281956%29.pdf
- Krathwohl, D. R. (2002) — *A Revision of Bloom's Taxonomy: An Overview*
  → Supports: explanation of the revised structure and its application to teaching, learning, and assessment
  → Why it matters: clarifies how the taxonomy connects learning objectives, activities, and assessment in practice
  → Source: https://doi.org/10.1207/s15430421tip4104_2

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